Chapter 2

Large Language Model & Reasoning

大语言模型的原理

大语言模型的训练

大语言模型的评估

大语言模型的推理

大语言模型的解释

1.1 Reasoning和Inference的区别

ChatGPT o1/o3/o4、DeepSeek R1、Gemini 2 Flash Thinking、Claude 3.7 Sonnet (Extended Thinking) ……

1+1=

word count: 2

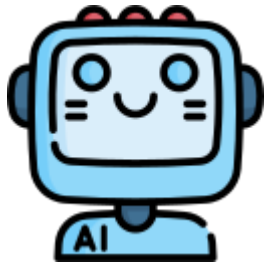
已深度思考 (5.8s)

好的，我现在需要弄清楚1加1等于多少。首先，我要理解每个数字代表什么。数字1就是一个单一的单位。比如，我有一个苹果，那就是一件水果。现在，如果我再加一个苹果，我总共有多少个苹果呢？

让我试着想象一下。假设我有一个苹果在桌子上。然后，有人给了我另一个苹果。现在，我一起数一数，我有一个，然后是另一个。所以，我总共有两个苹果。这是因为当我把两个单一的单位合并在一起时，我得到了两个单位的总数。

等等，有时候人们可能会把这个和其它运算混淆。比如，如果我有一个苹果组和另一个苹果组，这会改变结果吗？不会，因为每个组仍然有一个苹果，所以把它们合并在一起，我仍然得到两个苹果。这只是一个简单的加法过程。

1.1 Reasoning和Inference的区别

问题 →  → <think> 思考过程 </think> 答案

Verification

"Let me check the answer …"

Explore

"Let's try a different approach …"

Planning

"Let's first try to ……"

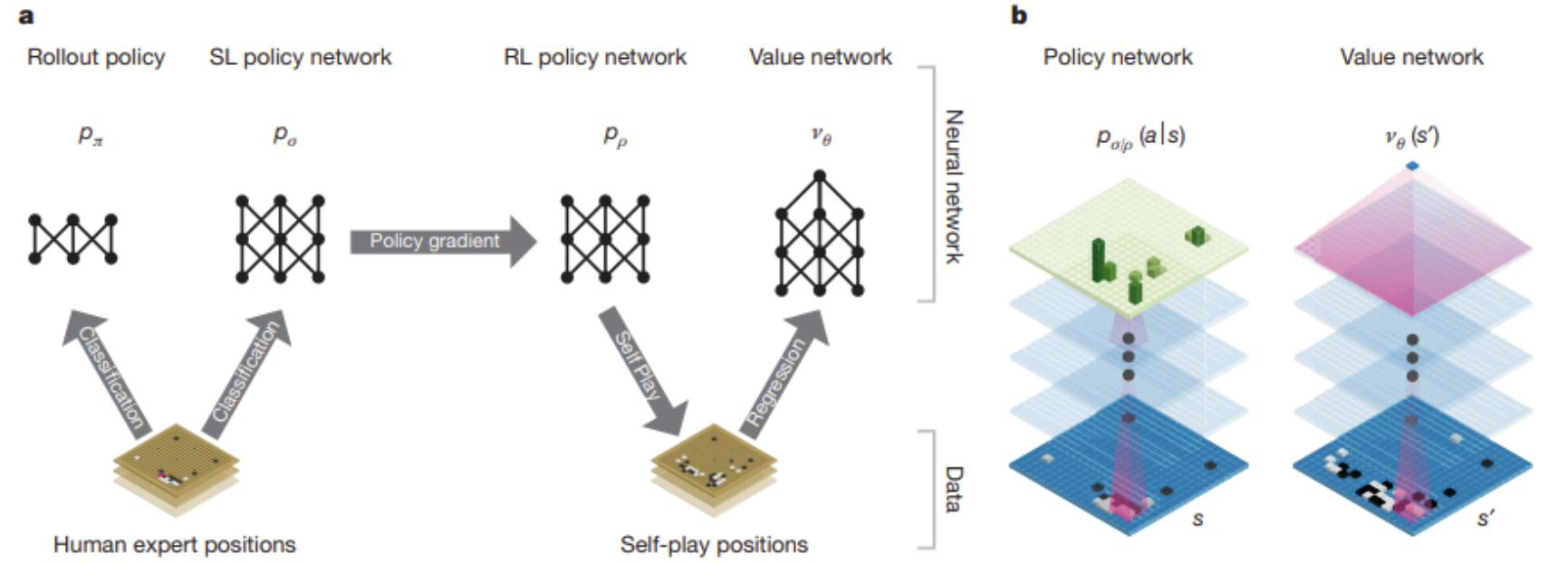
……

推理（Reasoning） （Inference 中文字面意思相近，意义完全不同）

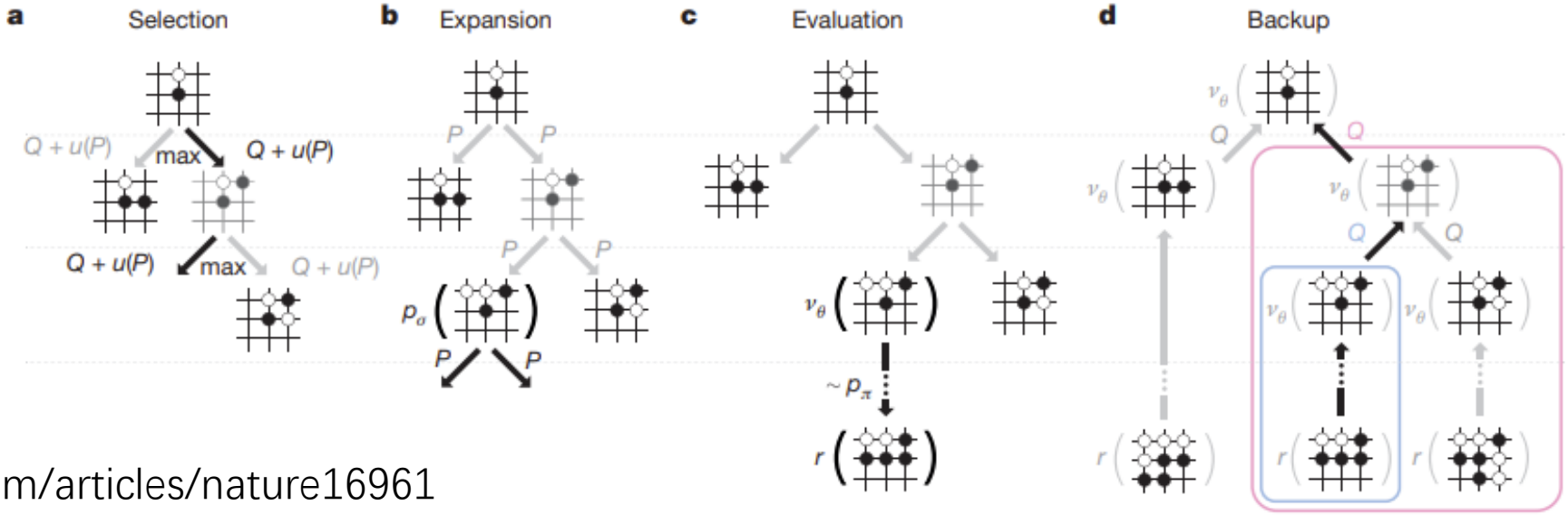
Test-Time Computing **"深度不够，长度来凑"**

1.2 Reasoning其实很早就有（AlphaGo）

Training Time

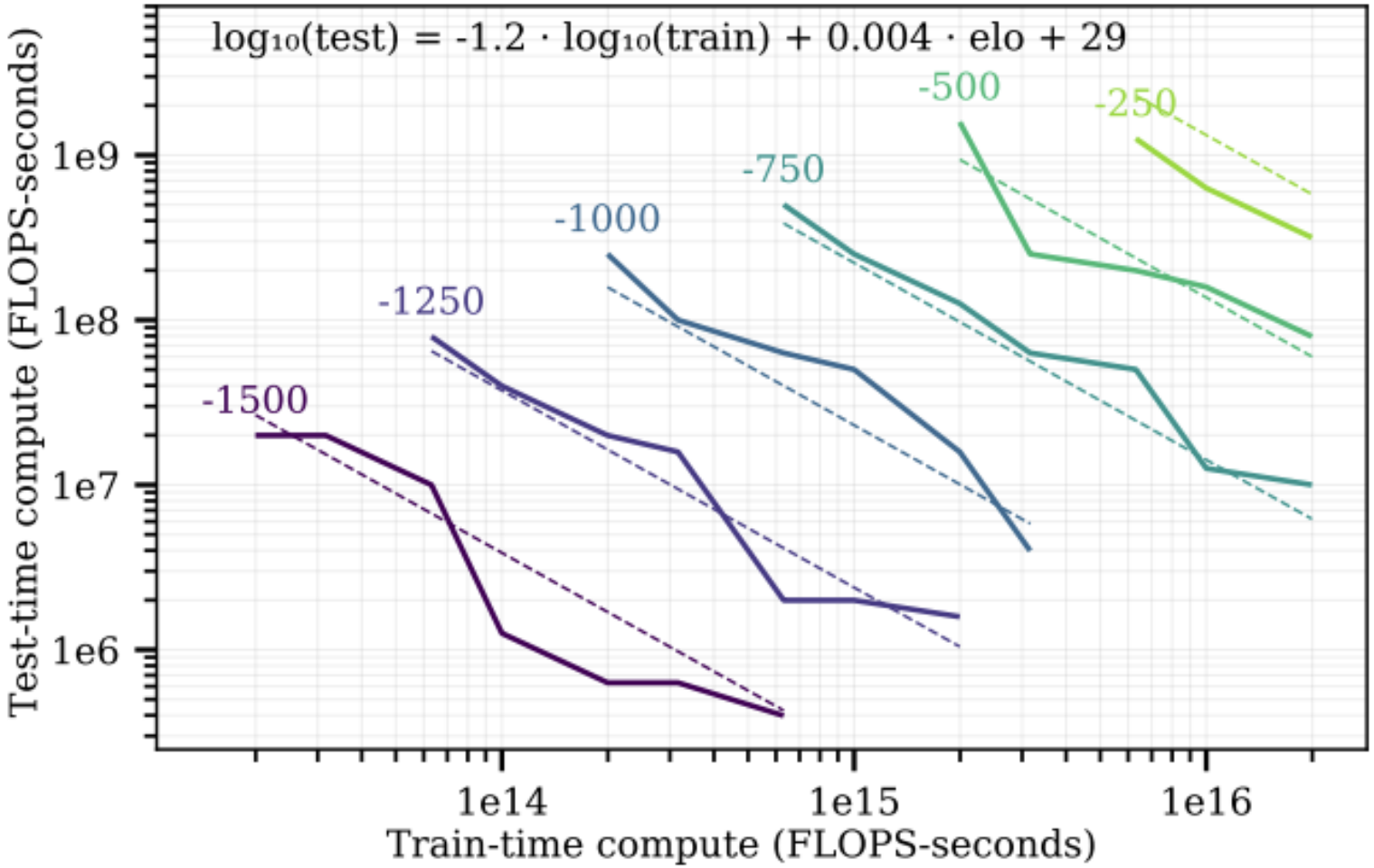


Testing Time



https://www.nature.com/articles/nature16961

1.3 Reasoning越多结果越好



Scaling Scaling Laws with Board Games
https://arxiv.org/abs/2104.03113

如何打造推理模型?

2.1 打造推理模型的方法

无需微调参数

更强的思维链（Chain-of-Thought, CoT）

设计模型推理的流程

教模型推理过程（Imitation Learning）

以结果导向的推理（Reinforcement Learning, RL）

需要微调参数

2.2 更强的思维链

更强的思维链（Chain-of-Thought, CoT）

设计模型推理的流程

教模型推理过程（Imitation Learning）

以结果导向的推理（Reinforcement Learning, RL）

2.2 更强的思维链

Few-shot CoT

https://arxiv.org/abs/2201.11903

"Short CoT"

Zero-shot CoT

https://arxiv.org/abs/2205.11916

推理模型在做的事：Long CoT

https://arxiv.org/abs/2503.09567

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?
A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A:

*(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are 16 / 2 = 8 golf balls. Half of the golf balls are blue. So there are 8 / 2 = 4 blue golf balls. **The answer is 4.*** ✓

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?
A: ***Let's think step by step.***

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

2.2 更强的思维链

请仔细思考并详细回答以下问题。在回答前，请先深入分析题目的要求，订出一个完整且清晰的解题计划，明确列出你将如何分步完成这个问题。在执行每一个主要步骤前，请再次制定该步骤的子计划，仔细列出需要处理的细节，然后再按部就班的执行。每执行完一个步骤或子步骤后，请进行多次验算，确保该步骤的答案绝对正确，并考量所有可能的解法。若在验算过程中发现问题，请立即回到该步骤重新制定或调整计划。在进行以上过程时，务必将你详细而完整的思考过程以及所有计划、子计划、验算步骤，全部置于"**<think>**"和"**</think>**"这两个标签符号之间。

123 x 456 =

Supervised CoT

https://arxiv.org/abs/2410.14198

2.2 更强的思维链

<think> 用户要求我仔细思考并详细回答123 x 456的计算问题。我需要制定一个完整的解题计划，并按照要求分步执行。

首先分析题目：这是一个三位数乘以三位数的乘法运算。我需要计算**123 × 456**。

解题计划：

1. 使用竖式乘法进行计算
2. 分步计算**123**与**456**中每一位的乘积
3. 将所有部分积相加得到最终结果

Supervised CoT

https://arxiv.org/abs/2410.14198

2.2 更强的思维链

让我制定详细的子计划：

步骤1：计算123 × 6

- 3 × 6 = 18，写下8，进位1
- 2 × 6 = 12，加上进位1 = 13，写下3，进位1
- 1 × 6 = 6，加上进位1 = 7
- 所以123 × 6 = 738

验算：123 × 6 = (100+20+3)×6 = 600+120+18 = 738 ✓

步骤2：计算123 × 50（实际上是123 × 5 × 10）

- 3 × 5 = 15，写下5，进位1
- 2 × 5 = 10，加上进位1 = 11，写下1，进位1
- 1 × 5 = 5，加上进位1 = 6
- 所以123 × 5 = 615
- 因为是乘以50，所以实际上是6150

验算：123 × 50 = (100+20+3)×50 = 5000+1000+150 = 6150 ✓

Supervised CoT

https://arxiv.org/abs/2410.14198

2.2 更强的思维链

步骤3：计算123 × 400（实际上是123 × 4 × 100）

- 3 × 4 = 12，写下2，进位1
- 2 × 4 = 8，加上进位1 = 9
- 1 × 4 = 4
- 所以123 × 4 = 492
- 因为是乘以400，所以实际上是49200

验算：123 × 400 = (100+20+3)×400 = 40000+8000+1200 = 49200 ✓

Supervised CoT

https://arxiv.org/abs/2410.14198

2.2 更强的思维链

步骤4：将所有部分积相加

- 738 (123 × 6)
- 6150 (123 × 50)
- 49200 (123 × 400)

竖式加法：

49200

6150

- 738

从右向左逐位相加：

- 个位：0+0+8 = 8
- 十位：0+5+3 = 8
- 百位：2+1+7 = 10，写下0，进位1
- 千位：9+6+进位1 = 16，写下6，进位1
- 万位：4+进位1 = 5

所以结果是56088

验算：738 + 6150 = 6888，6888 + 49200 = 56088 ✓

Supervised CoT

https://arxiv.org/abs/2410.14198

2.2 更强的思维链

我还可以用另一种方法验证：

123 × 456 = 123 × (400 + 50 + 6) = 123×400 + 123×50 + 123×6

= 49200 + 6150 + 738 = 56088 ✓

所有计算都一致，确认结果是正确的。

</think>

最终输出

123 × 456 = 56088

计算过程如下：

- **123 × 6 = 738**
- **123 × 50 = 6,150**
- **123 × 400 = 49,200**
- 将三个部分积相加：**738 + 6,150 + 49,200 = 56,088**

经过多次验算，确认结果正确无误。

不是所有大语言模型都有能力根据复杂指令进行Long CoT

Supervised CoT

https://arxiv.org/abs/2410.14198

2.3 设计模型推理的流程

更强的思维链（Chain-of-Thought, CoT）

设计模型推理的流程

教模型推理过程（Imitation Learning）

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2.3.1 能力弱的模型也能推理

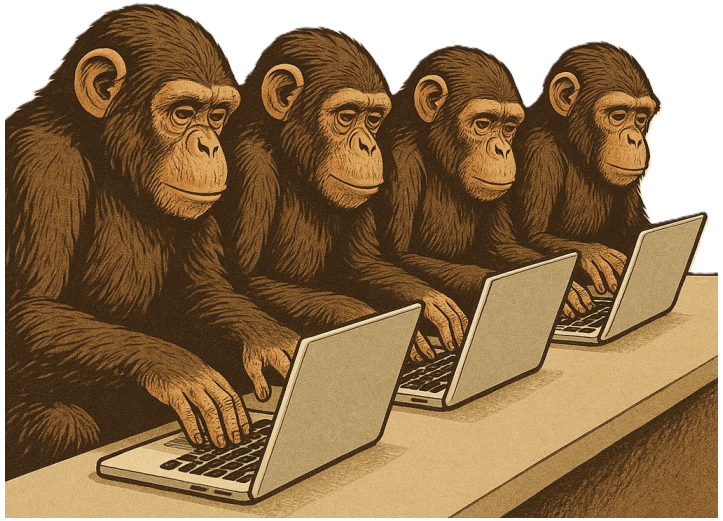
无限猴子定理
Infinite Monkey Theorem

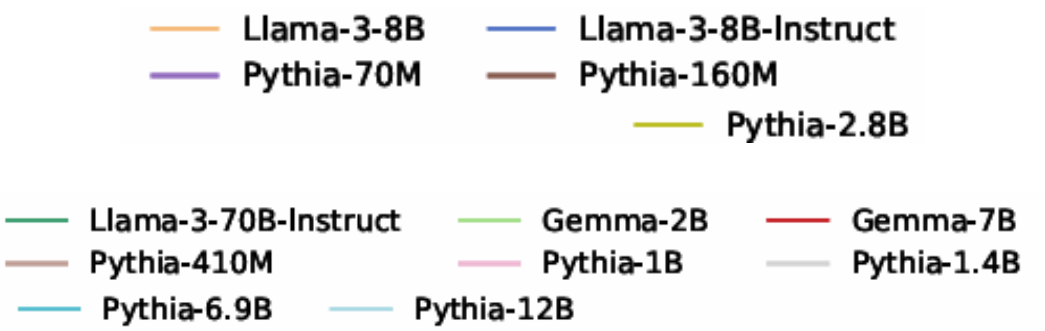
让一只猴子在打字机上随机地按键，当按键时间达到无穷时，几乎必然能够打出任何给定的文字，比如莎士比亚的全套著作。

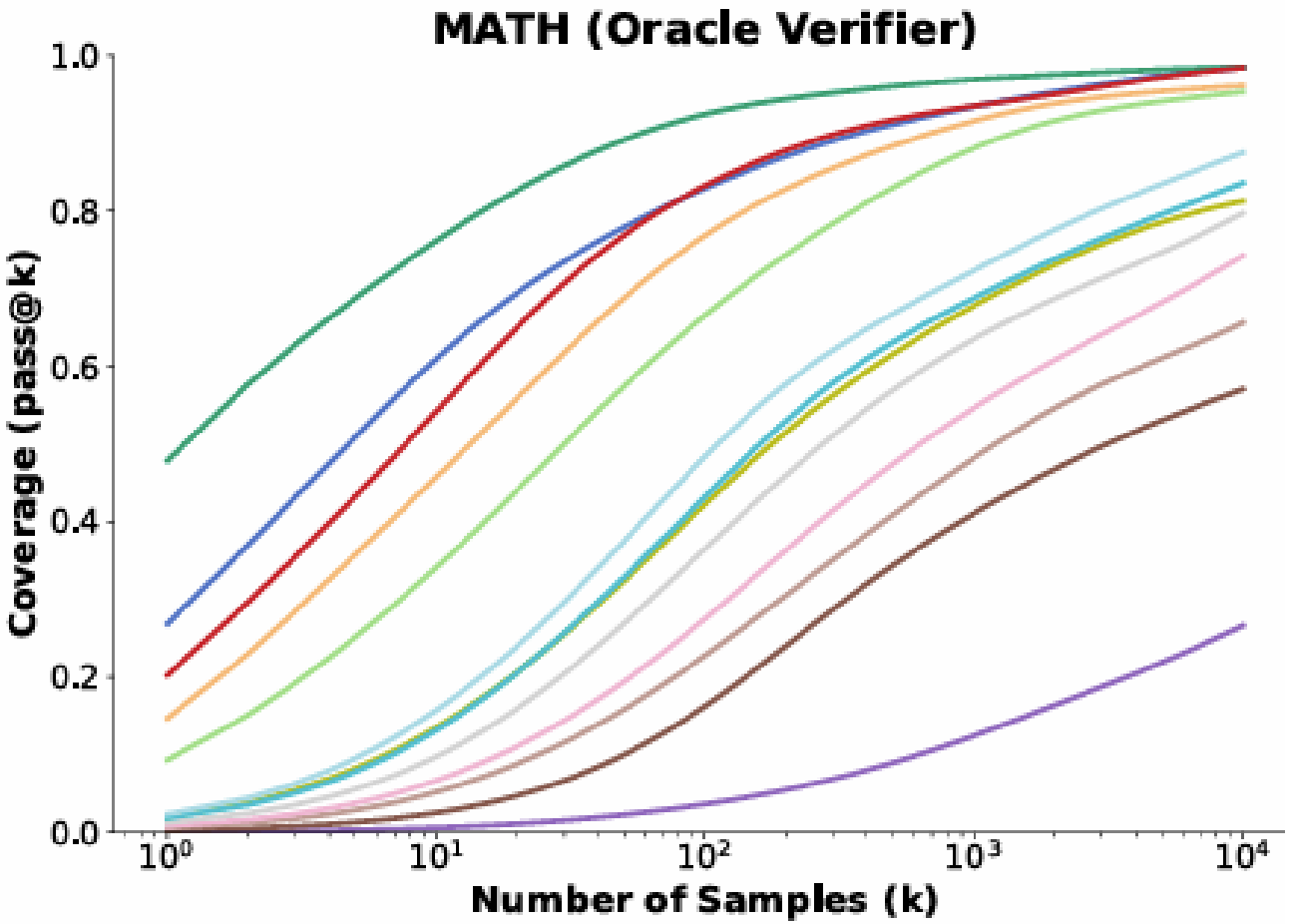
2.3.1 能力弱的模型也能推理

“Large Language Monkeys”

https://arxiv.org/abs/2407.21787



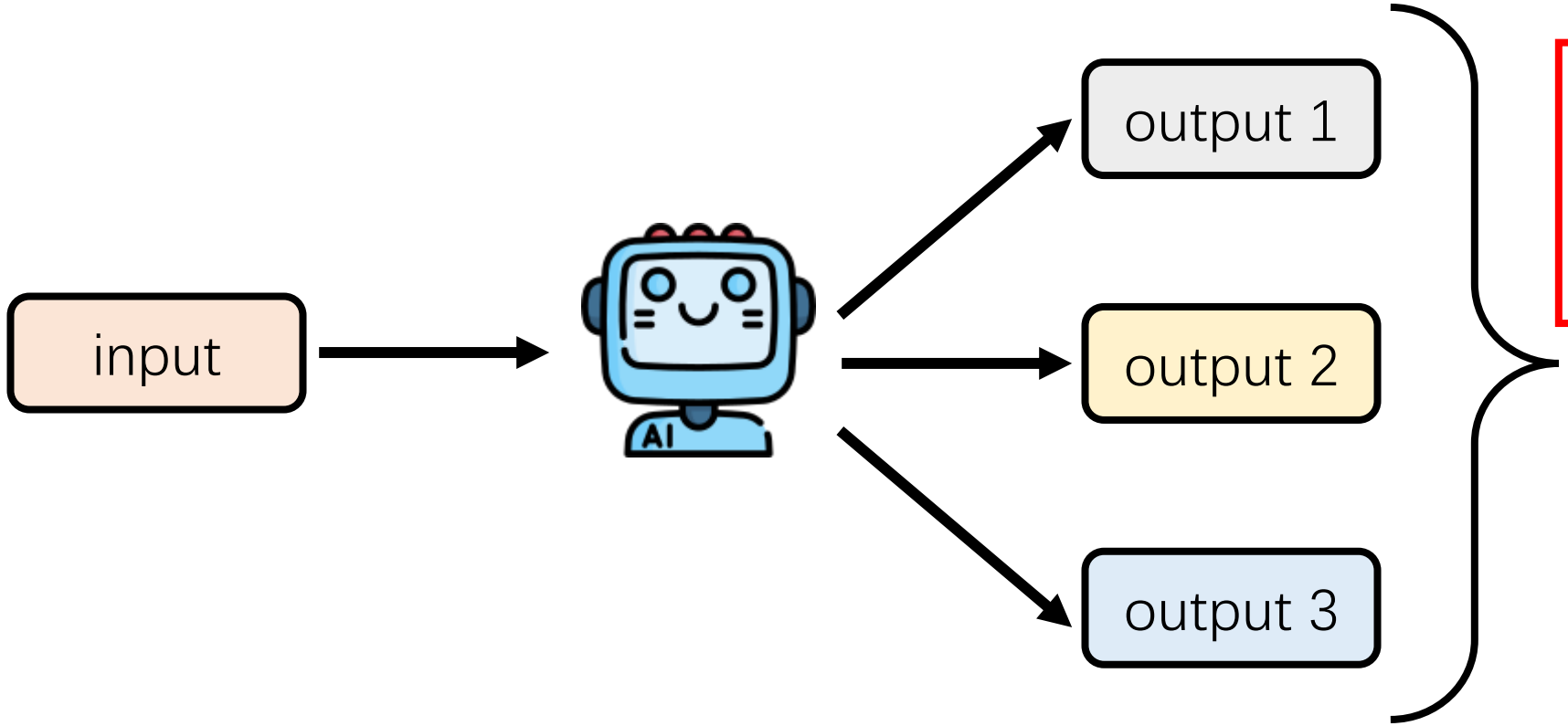




2.3.2 Majority Vote

- 让LLMs多试几次

如何选出正确的答案？

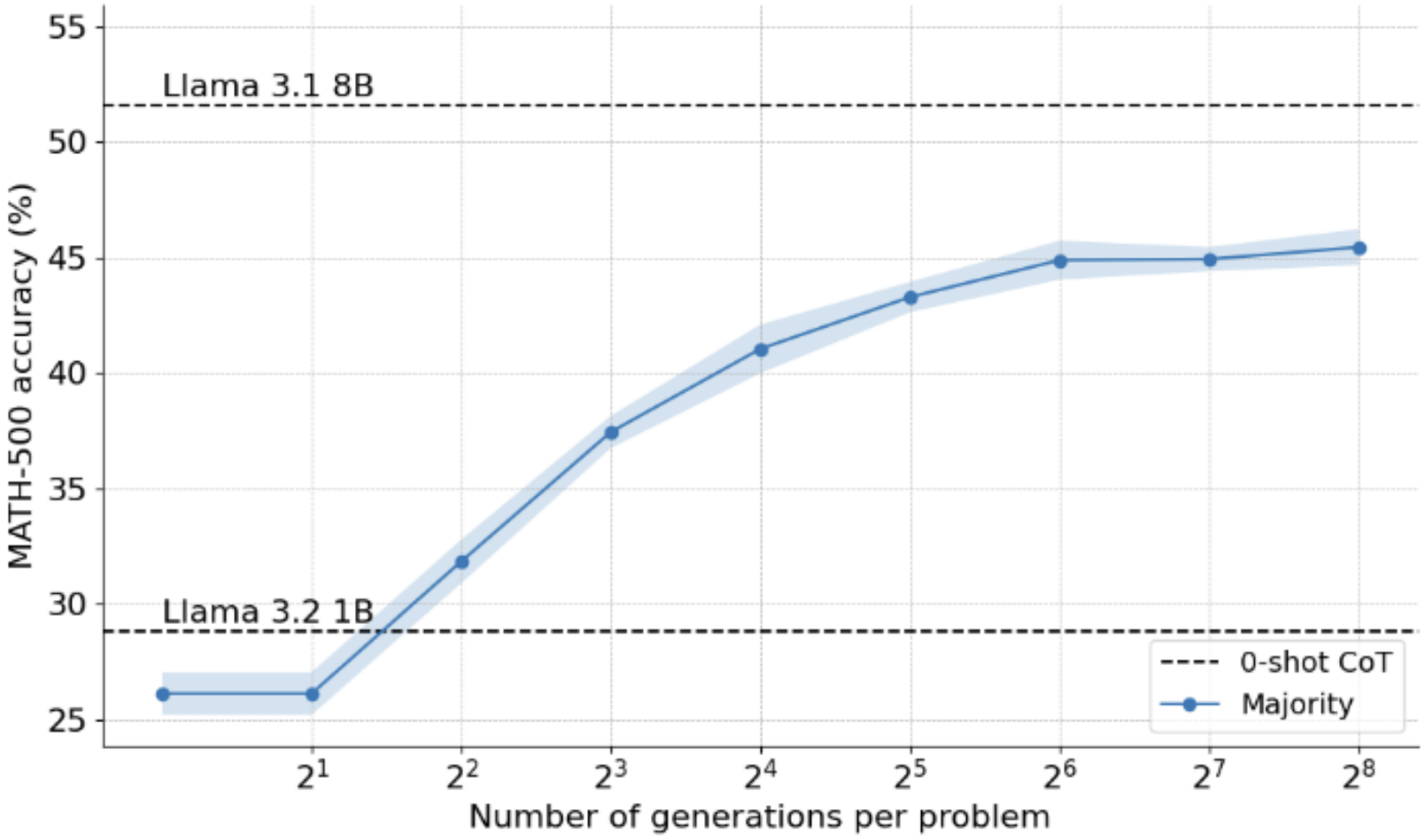


Majority Vote
(Self-consistency)
https://arxiv.org/abs/2203.11171

<answer>答案</answer>

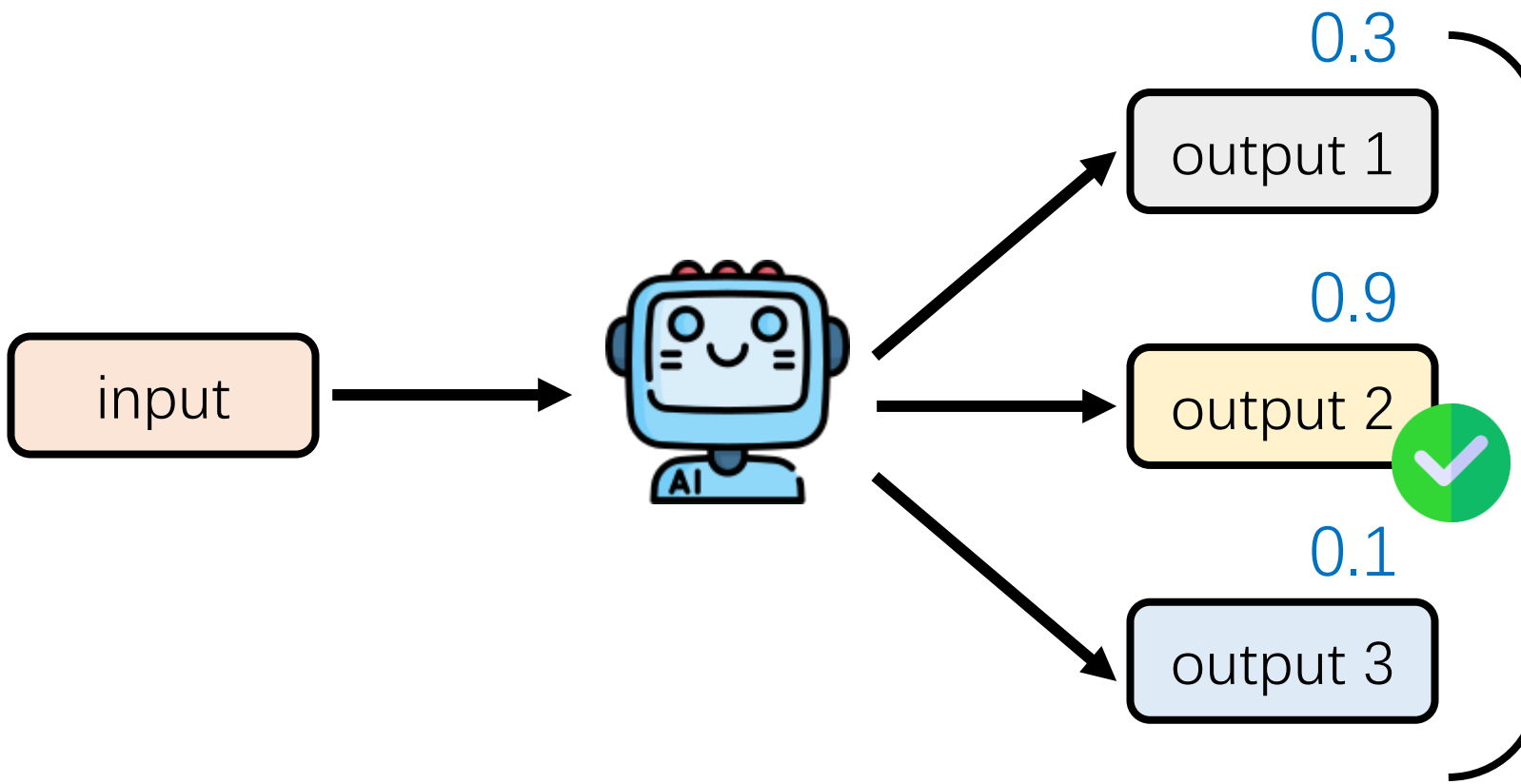
Confidence
(used in CoT decoding)

2.3.2 Majority Vote



https://huggingface.co/spaces/HuggingFaceH4/blogpost-scaling-test-time-compute

2.3.3 Output Verification



如何选出正确的答案？

Majority Vote
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Confidence
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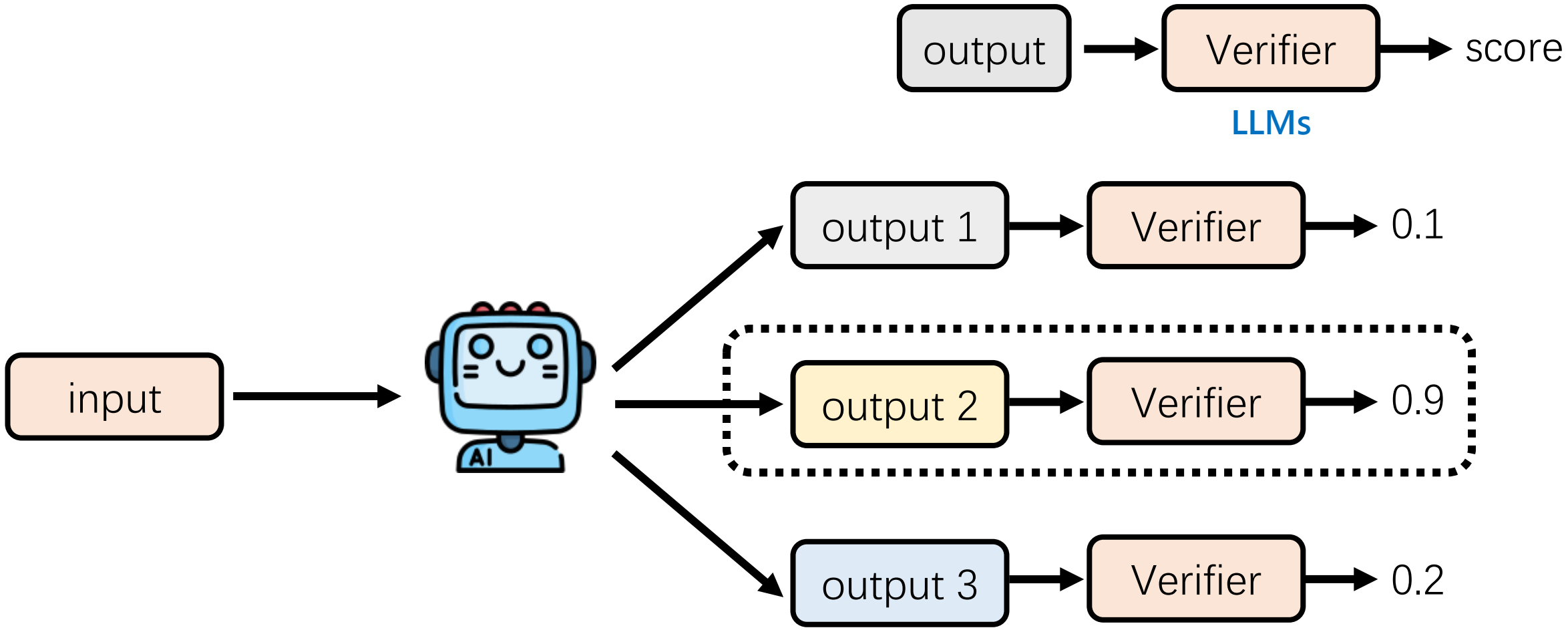
2.3.3 Output Verification



Best-of-N

https://arxiv.org/abs/2110.14168

但是，模型在推理时有可能出现错误积累

2.3.4 Step Verification

- 对步骤进行验证

计算123*456

word count: 4

已深度思考 (1m 25s)

首先，用户要求计算123乘以456。这是一个简单的乘法问题，我需要准确地计算它。

让我回想一下乘法的基本步骤：将两个数相乘，我可以使用长乘法的方法。或者，我可以直接计算，因为数字不大。

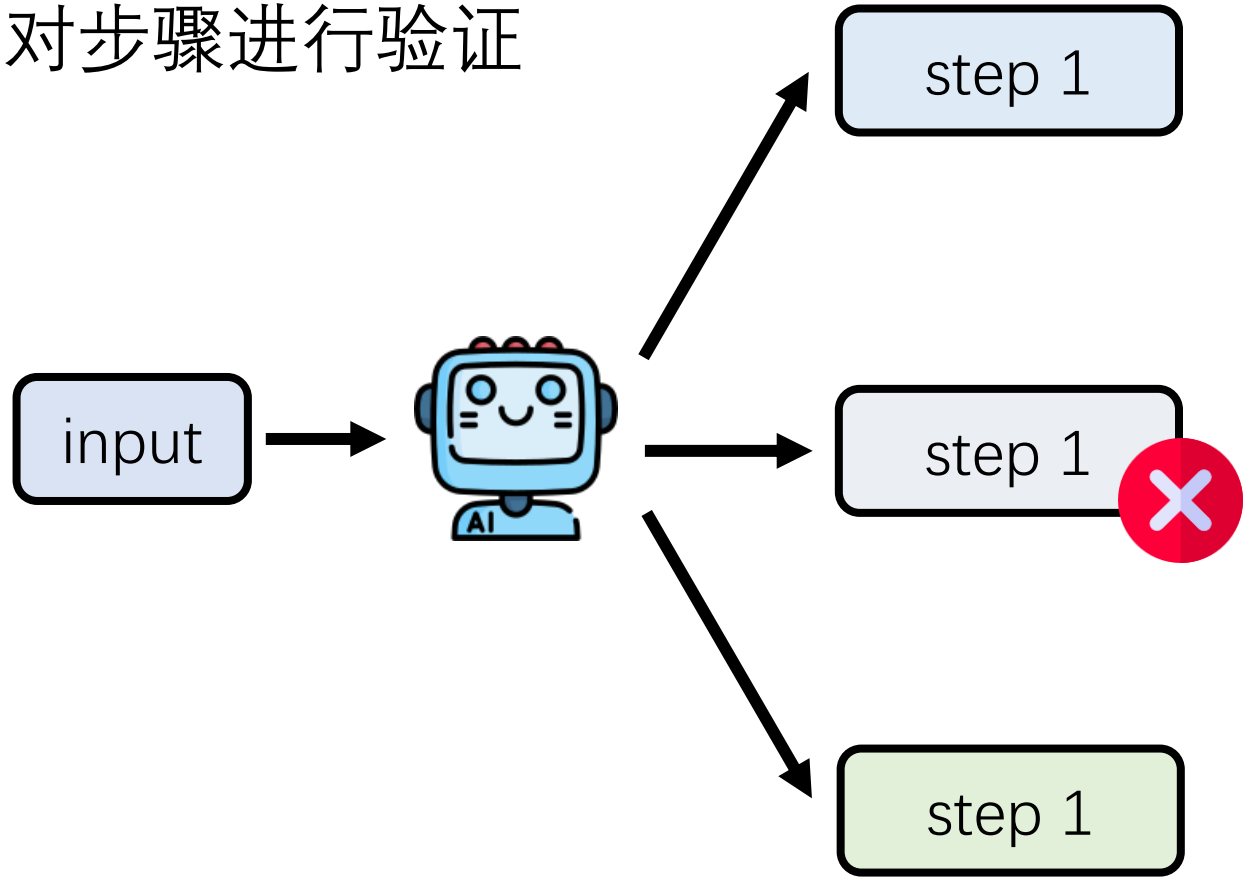
123乘以456。我可以分解它来更容易计算。

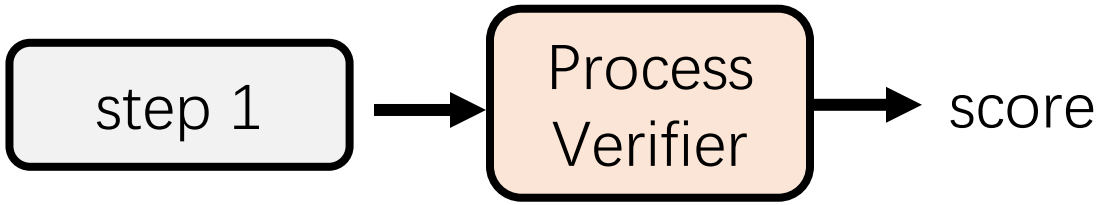
例如，123 * 456 = 123 * (400 + 50 + 6) = 123*400 + 123*50 + 123*6。

让我计算每个部分：

2.3.4 Step Verification

- 对步骤进行验证





Let's Verify Step by Step
https://arxiv.org/abs/2305.20050

2.3.4 Step Verification

请逐步解决问题，每一步骤以
<step>开始，以</step>结束

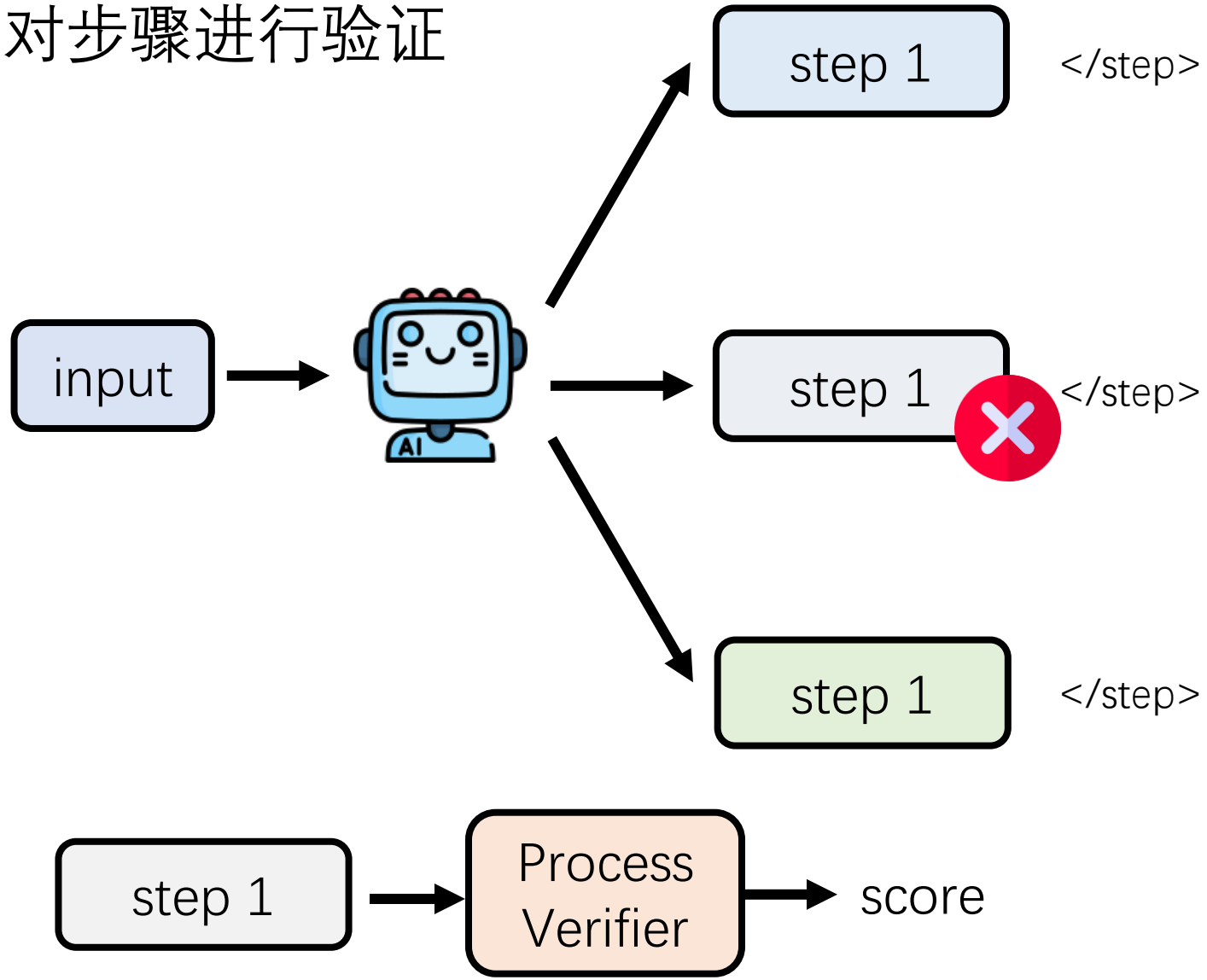
案例：
<step>步骤一</step>
<step>步骤二</step>
<answer>答案</answer>

问题：
123 x 456 =

<step>首先，将乘法分解为三个部分：**123 × 6**、**123 × 50** 和 **123 × 400**，然后相加。</step>

2.3.4 Step Verification

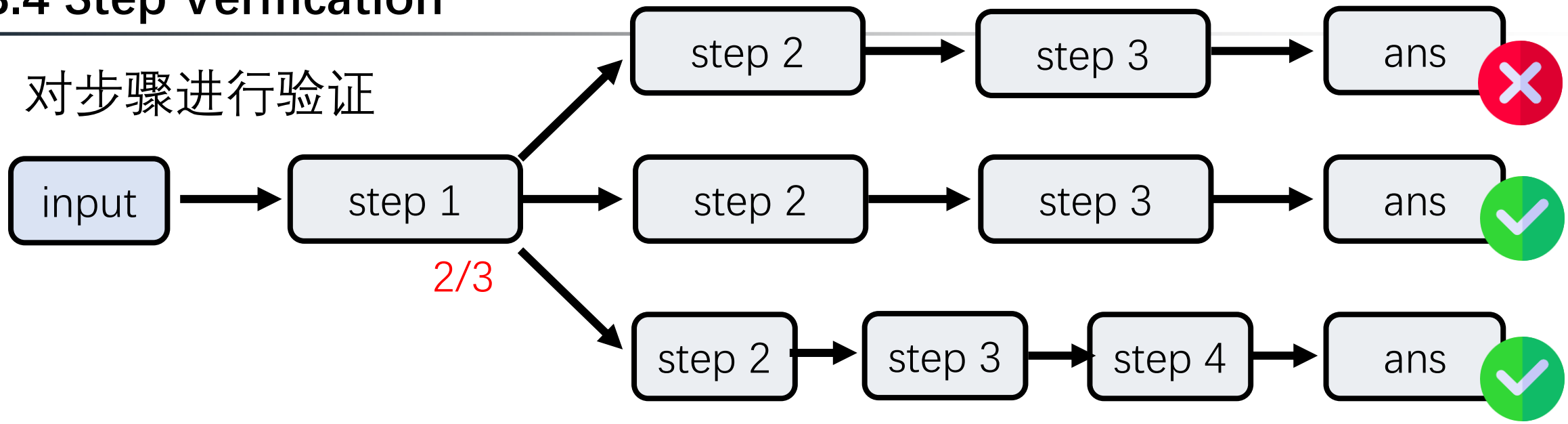
- 对步骤进行验证

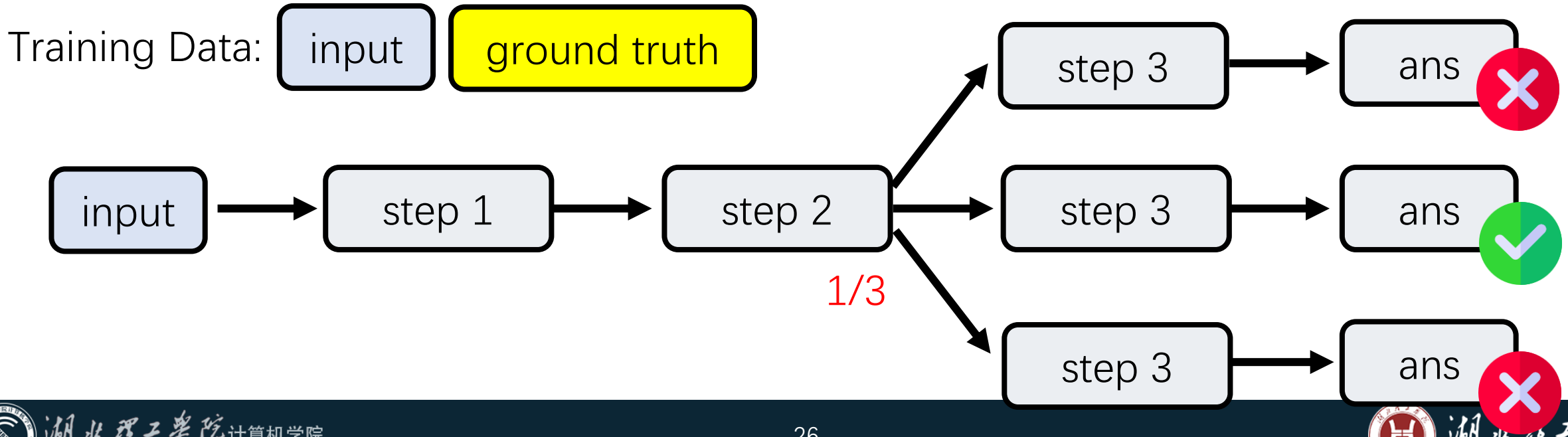


Let's Verify Step by Step
https://arxiv.org/abs/2305.20050

2.3.4 Step Verification

- 对步骤进行验证

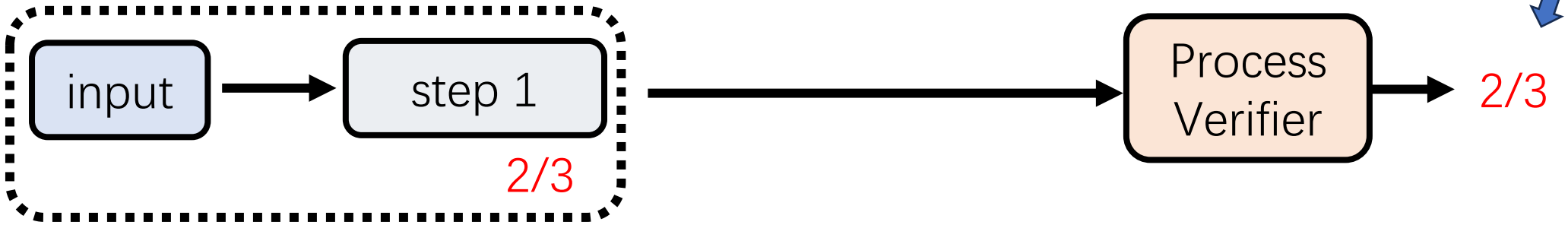




2.3.4 Step Verification

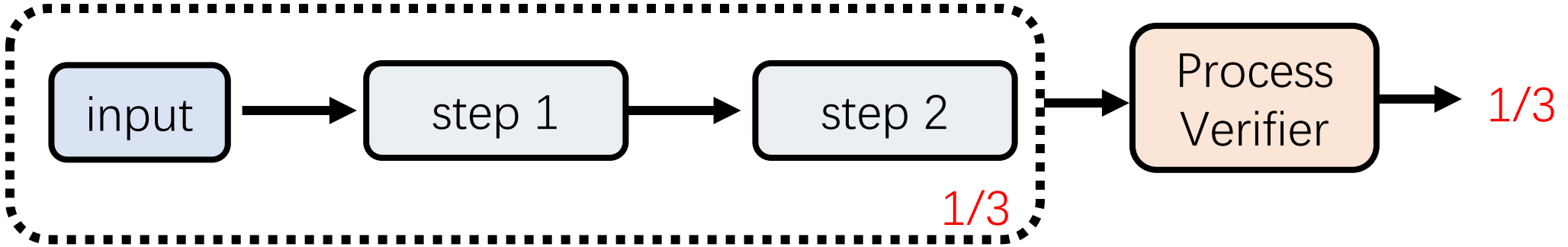
- 对步骤进行验证

超过多少概率算“好”?



Math-Shepherd: Verify and Reinforce LLMs Step-by-step without Human Annotations

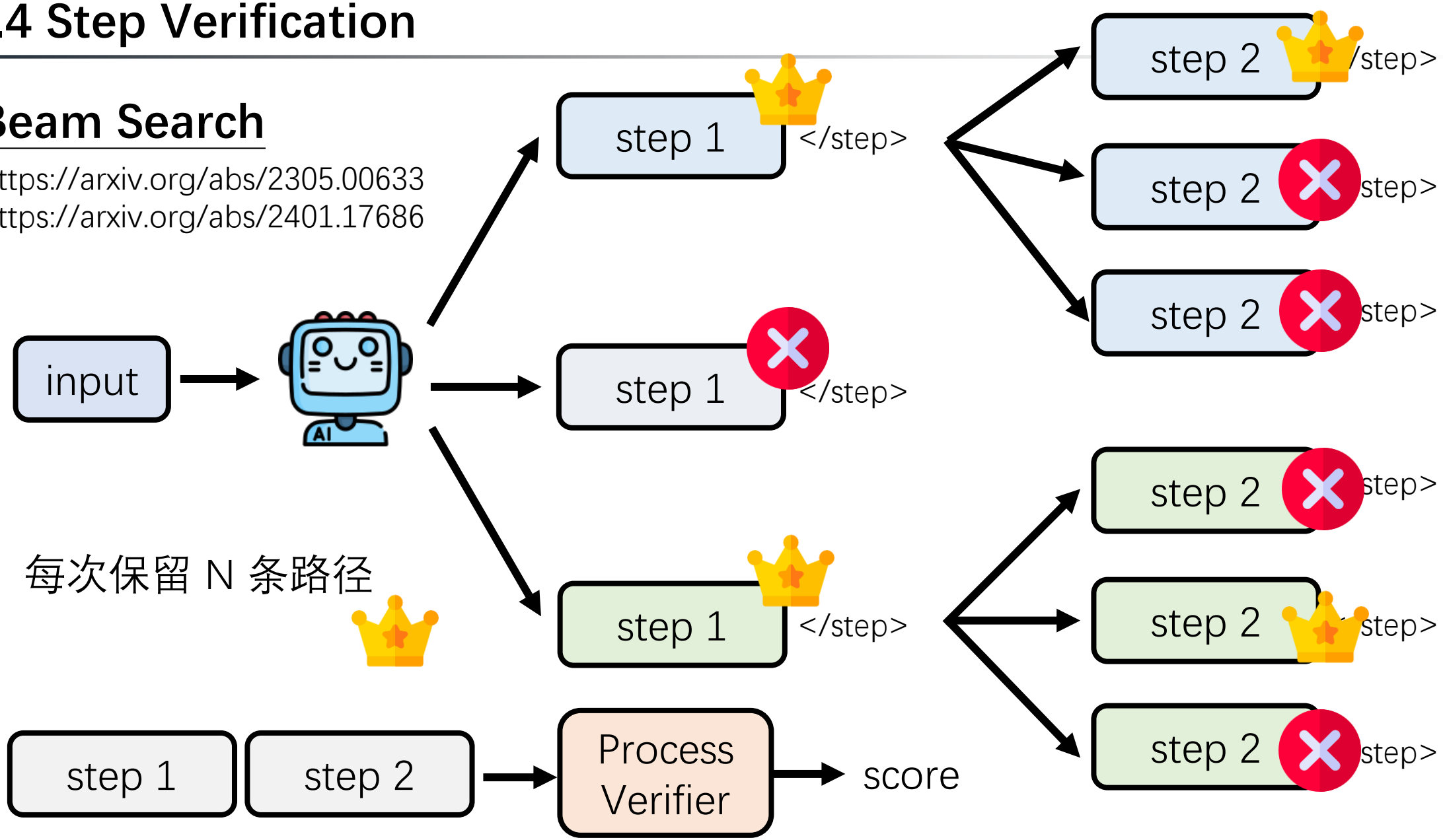
https://arxiv.org/abs/2312.08935



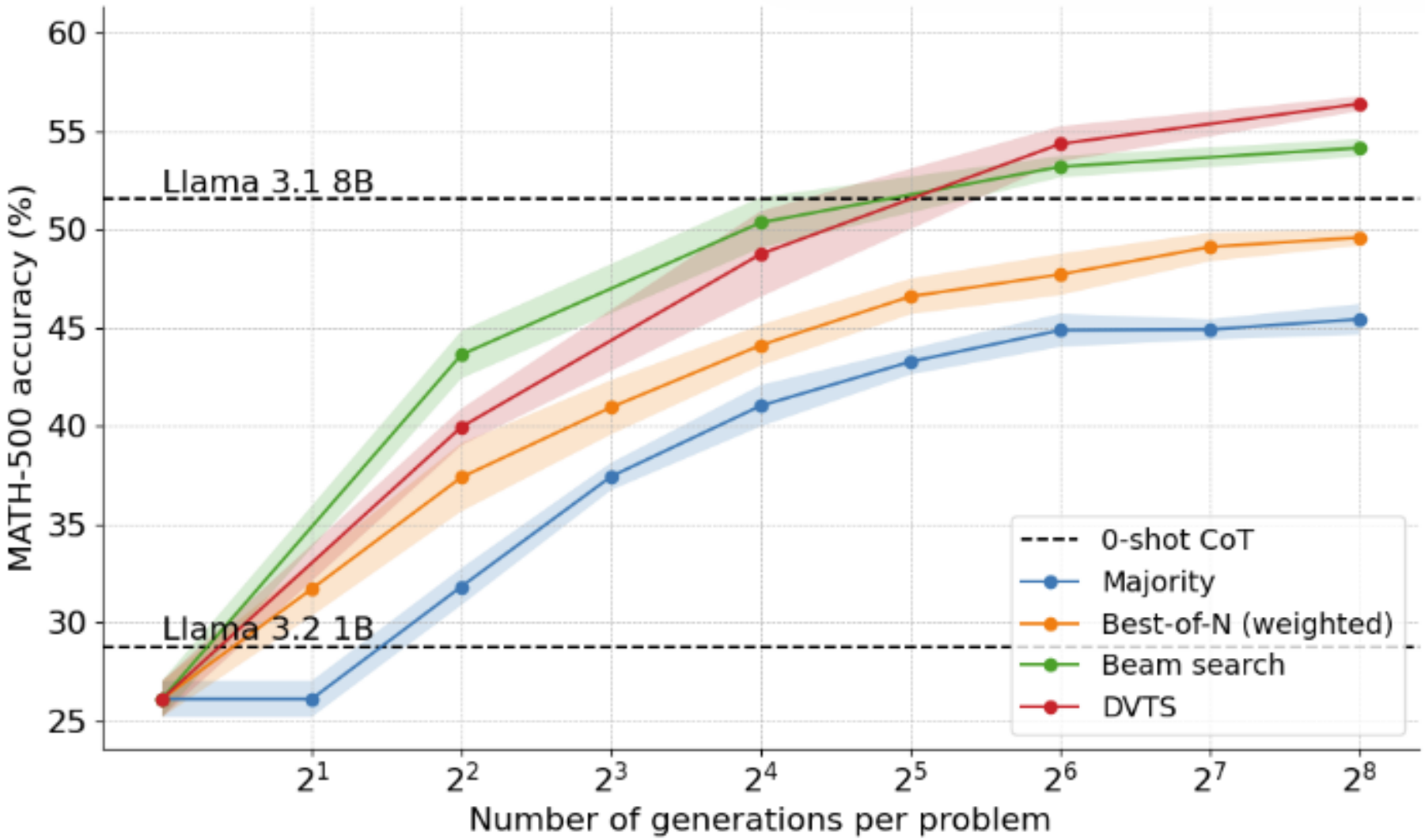
2.3.4 Step Verification

Beam Search

https://arxiv.org/abs/2305.00633
https://arxiv.org/abs/2401.17686



2.3.4 Step Verification

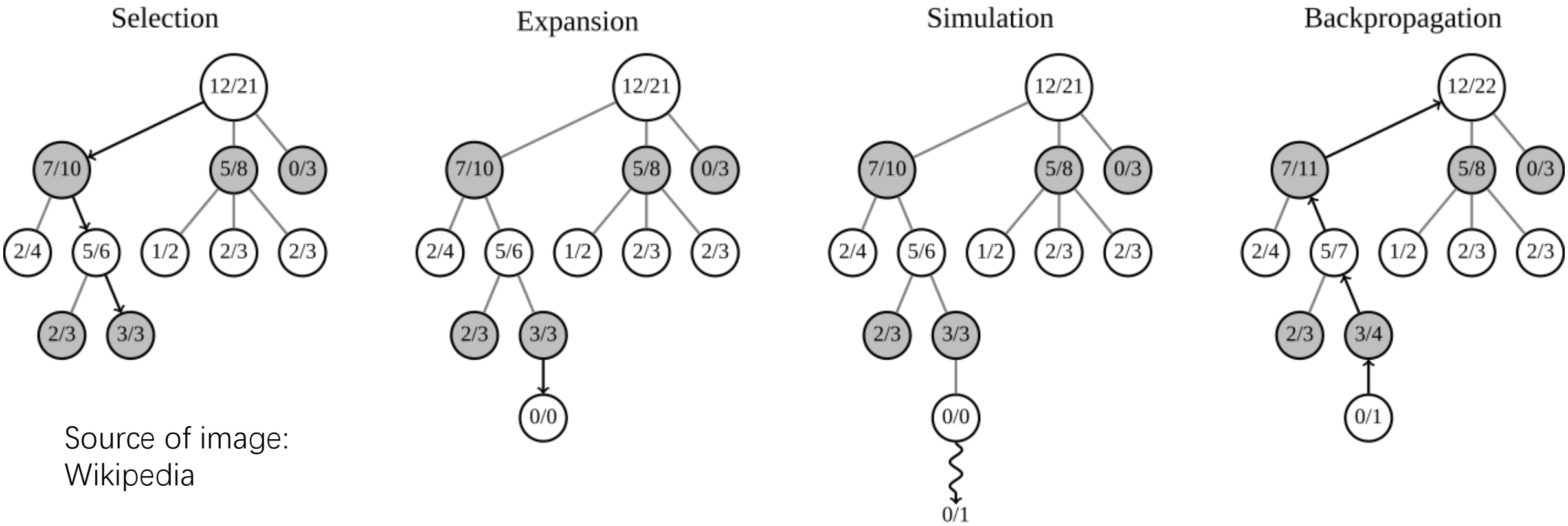


https://huggingface.co/spaces/HuggingFaceH4/blogpost-scaling-test-time-compute

2.3.4 Step Verification

Heuristic Search Algorithm

e.g. Monte Carlo Tree Search (MCTS)



Source of image: Wikipedia

Monte Carlo Tree Search Boosts Reasoning via Iterative Preference Learning https://arxiv.org/abs/2405.00451

ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search https://arxiv.org/abs/2406.03816

Mutual Reasoning Makes Smaller LLMs Stronger Problem-Solvers https://arxiv.org/abs/2408.06195

2.4 训练推理模型

LLM without Reasoning

(Foundation Model)

Post-Training

Learn to Reasoning

LLM with Reasoning

(Fine-tuned Model)

教模型推理过程（Imitation Learning）

以结果导向的推理（Reinforcement Learining, RL）

2.4 教模型推理过程

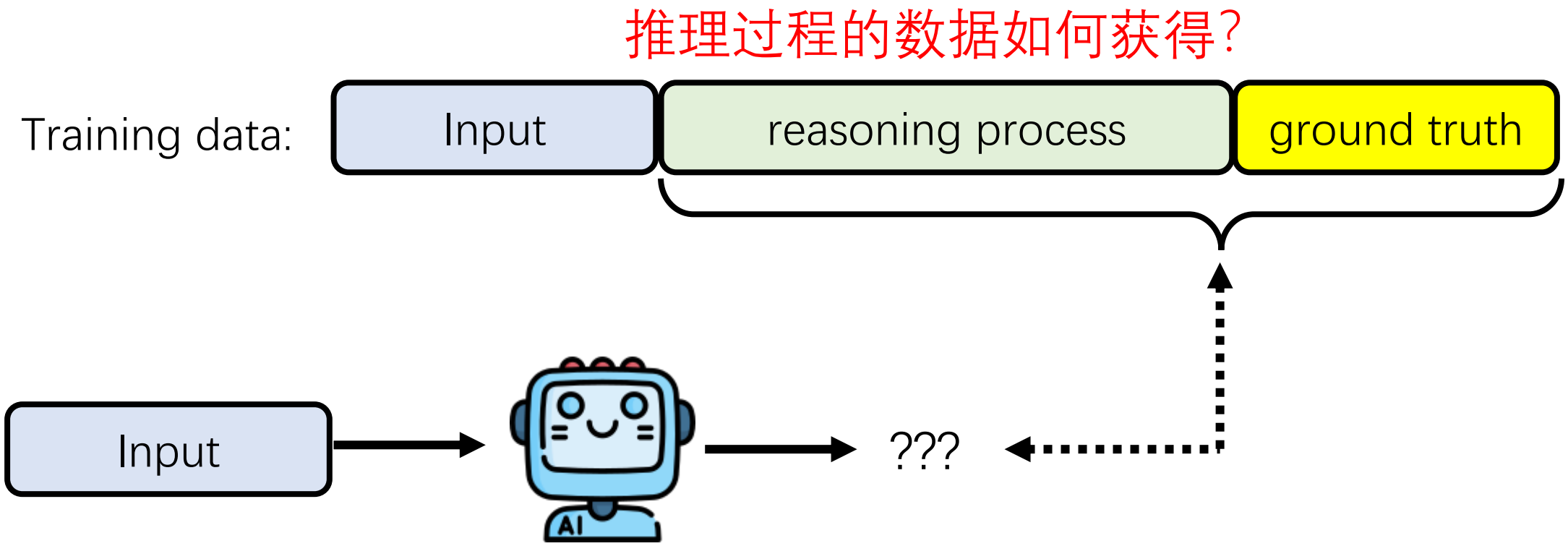
更强的思维链（Chain-of-Thought, CoT）

设计模型推理的流程

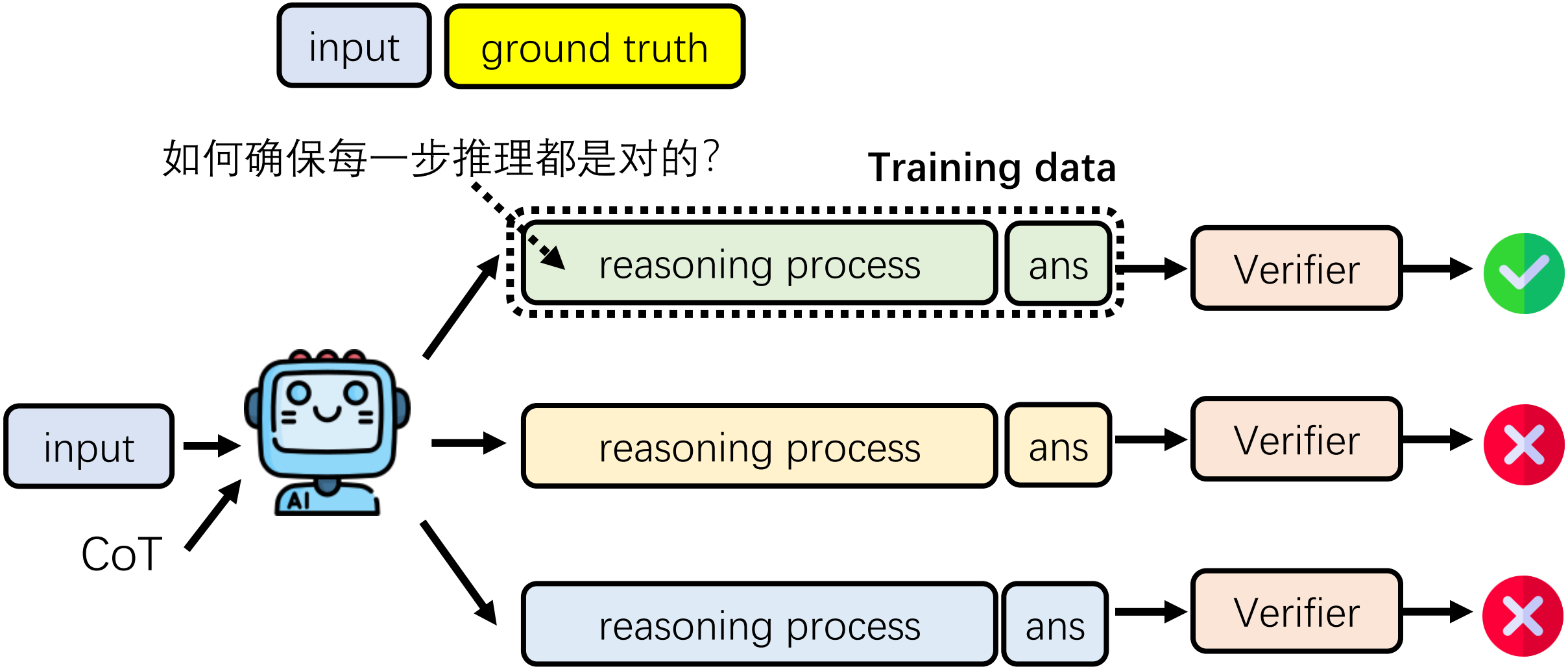
教模型推理过程（Imitation Learning）

以结果导向的推理（Reinforcement Learning, RL）

2.4 教模型推理过程


推理过程的数据如何获得?
Training data:
Input
reasoning process
ground truth
Input
AI
???

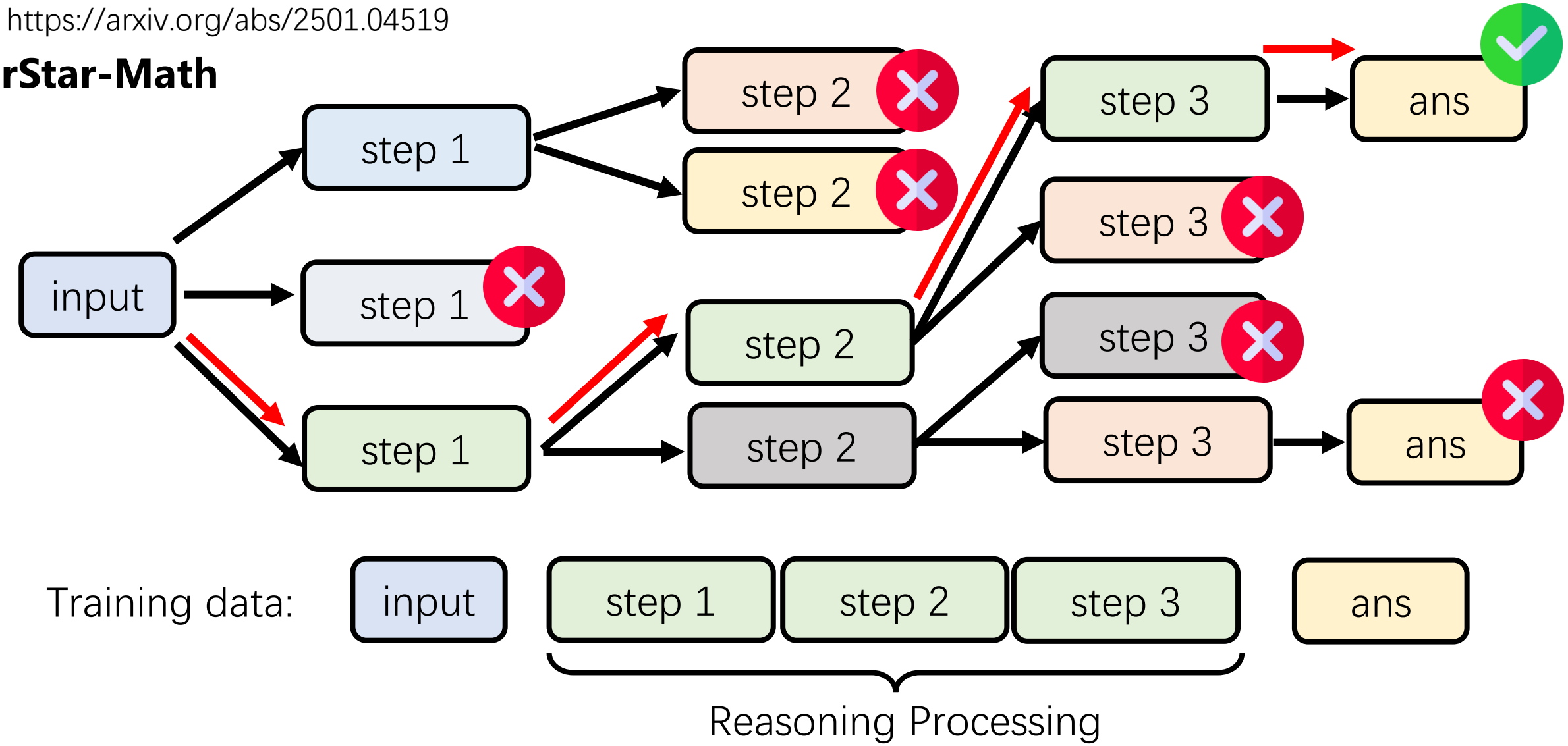
2.4.1 生成推理过程的数据



2.4.1 生成推理过程的数据

https://arxiv.org/abs/2501.04519

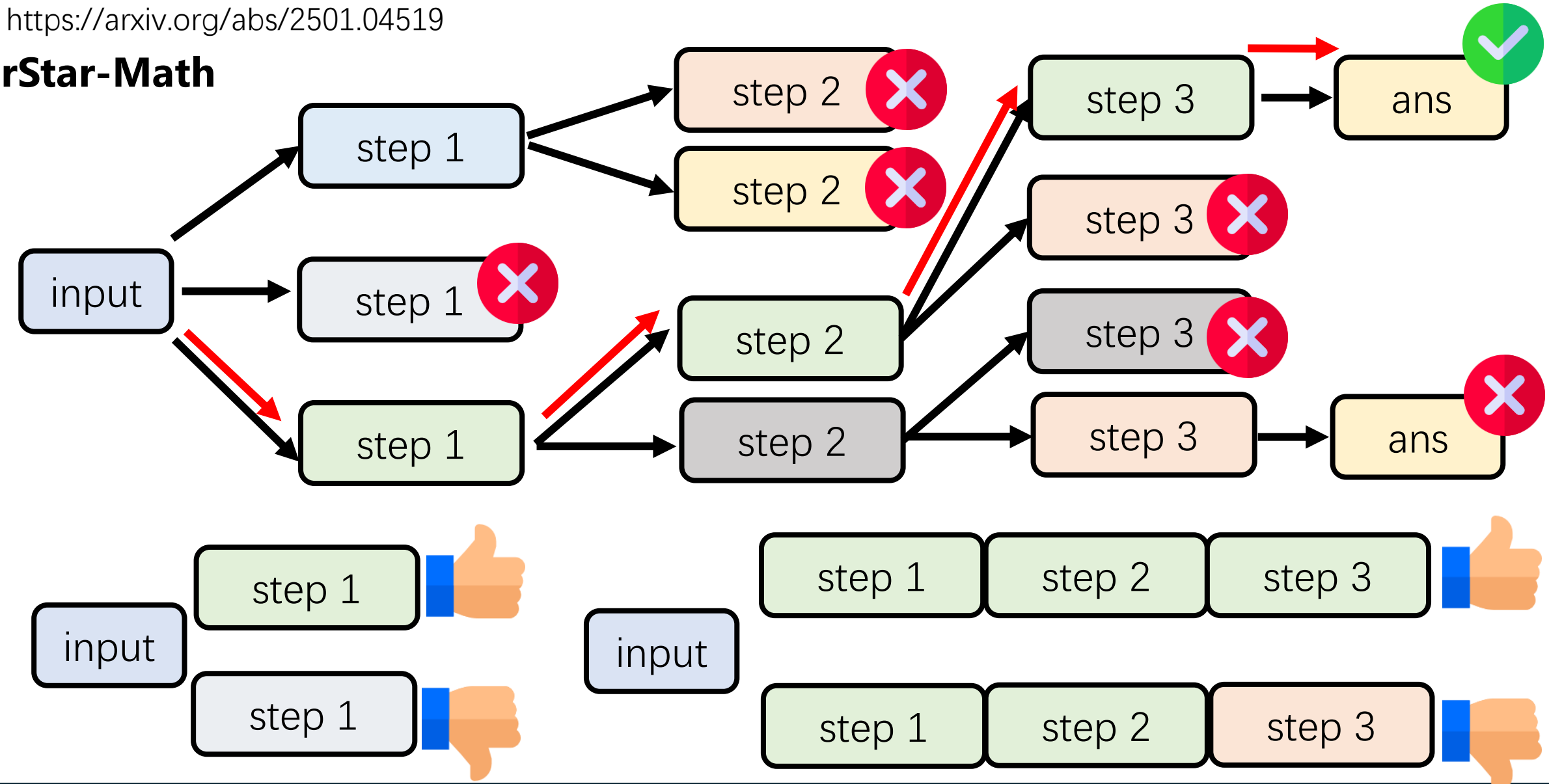
rStar-Math



2.4.1 生成推理过程的数据

https://arxiv.org/abs/2501.04519

rStar-Math



2.4.2 生成推理过程的每一步都必须正确吗？

计算123*456

word count: 4

已深度思考 (1m 25s)

首先，用户要求计算123乘以456。这是一个简单的乘法问题，我需要准确地计算它。

让我回想一下乘法的基本步骤：将两个数相乘，我可以使用长乘法的方法。或者，我可以直接计算，因为数字不大。

123乘以456。我可以分解它来更容易计算。

⋮

所以6888？但这不对，因为之前我算的是56088。我忘了还有百位。

⋮

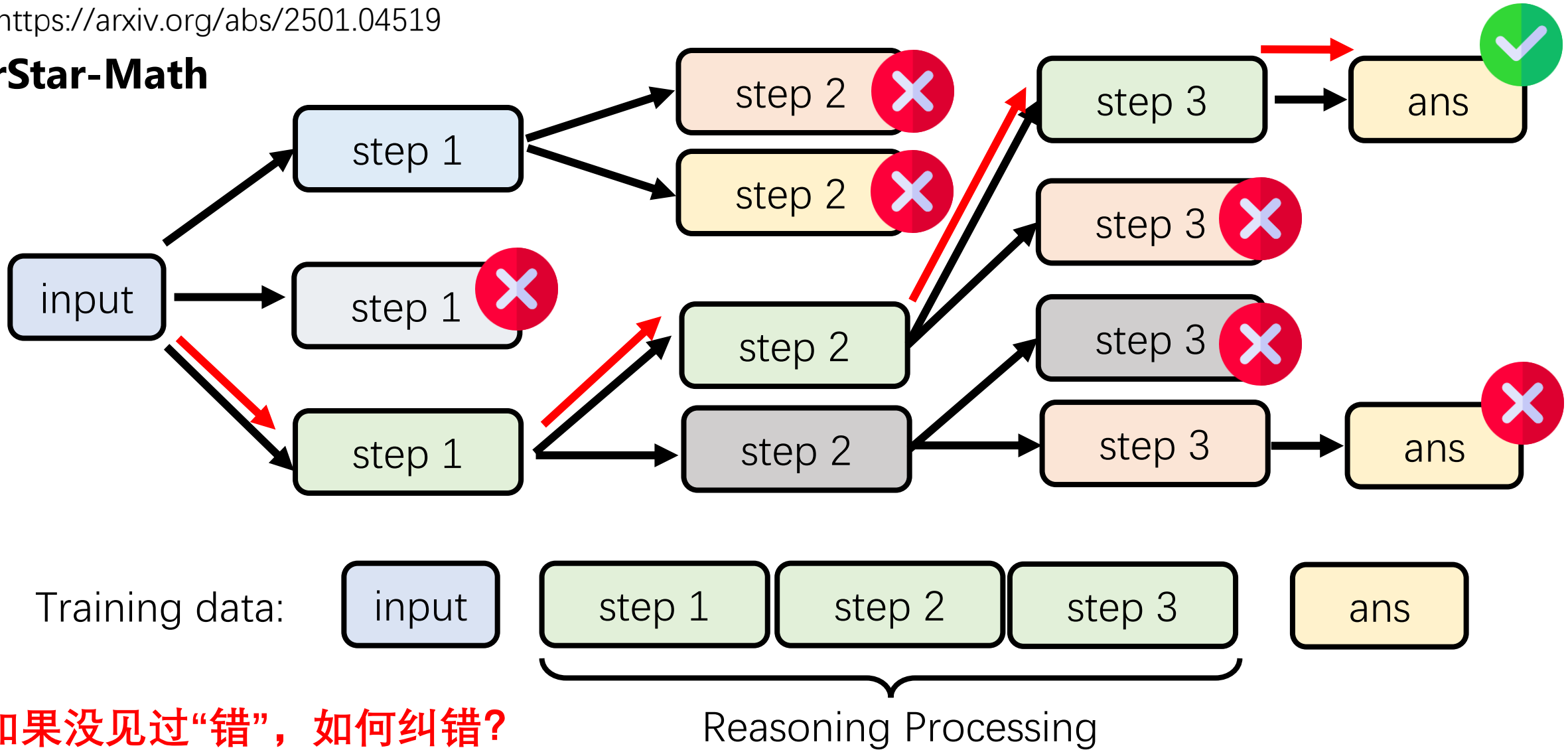
推理模型拥有纠错的能力

如果没见过“错”，如何纠错？

2.4.2 生成推理过程的每一步都必须正确吗？

https://arxiv.org/abs/2501.04519

rStar-Math

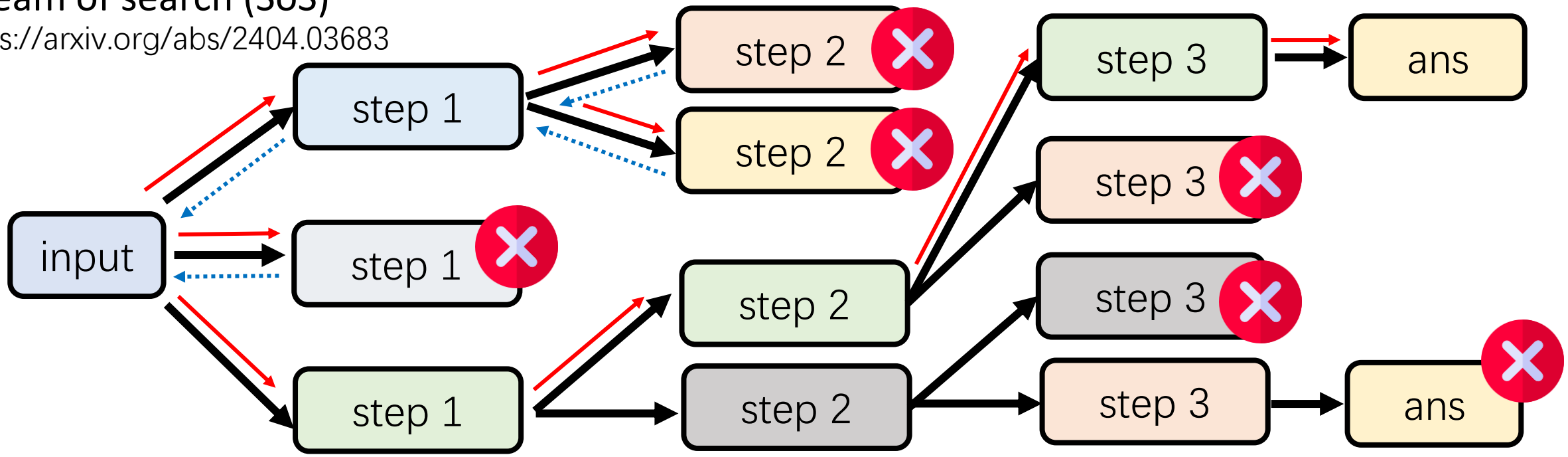


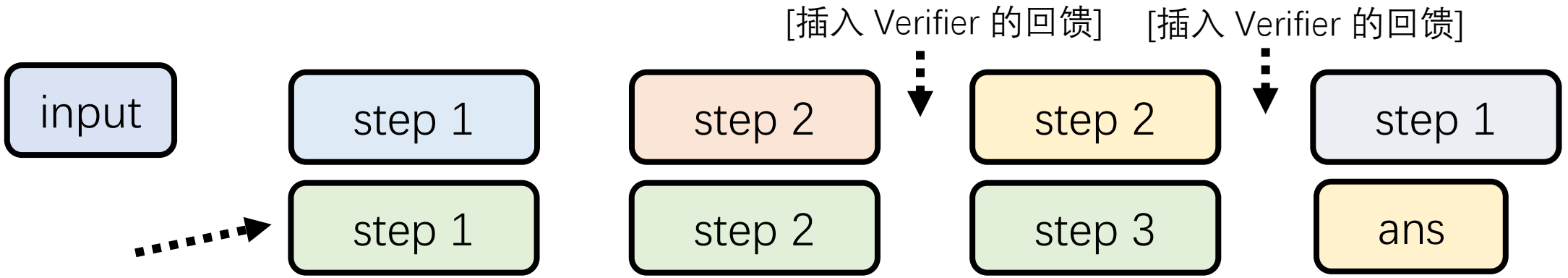
如果没见过“错”，如何纠错？

2.4.2 生成推理过程的每一步都必须正确吗？

Stream of search (SoS)

https://arxiv.org/abs/2404.03683

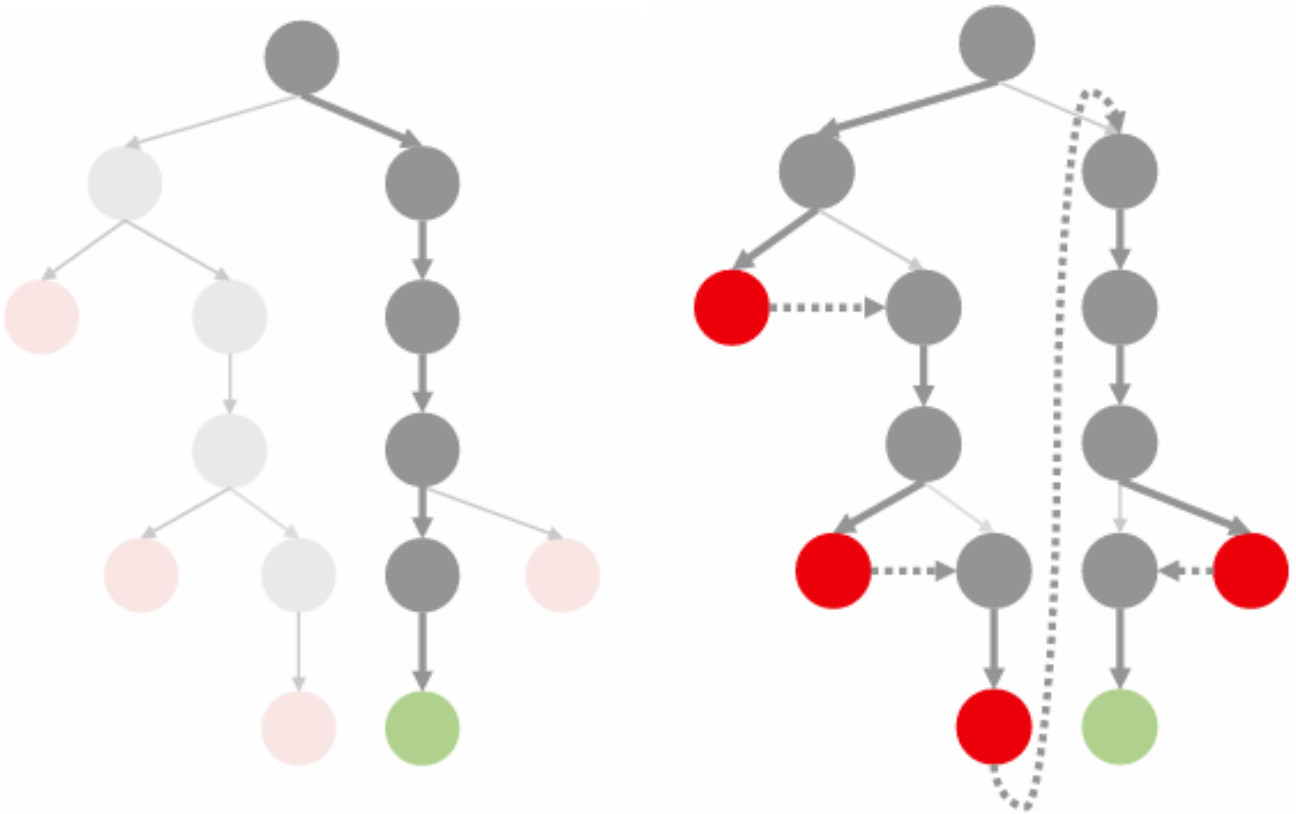




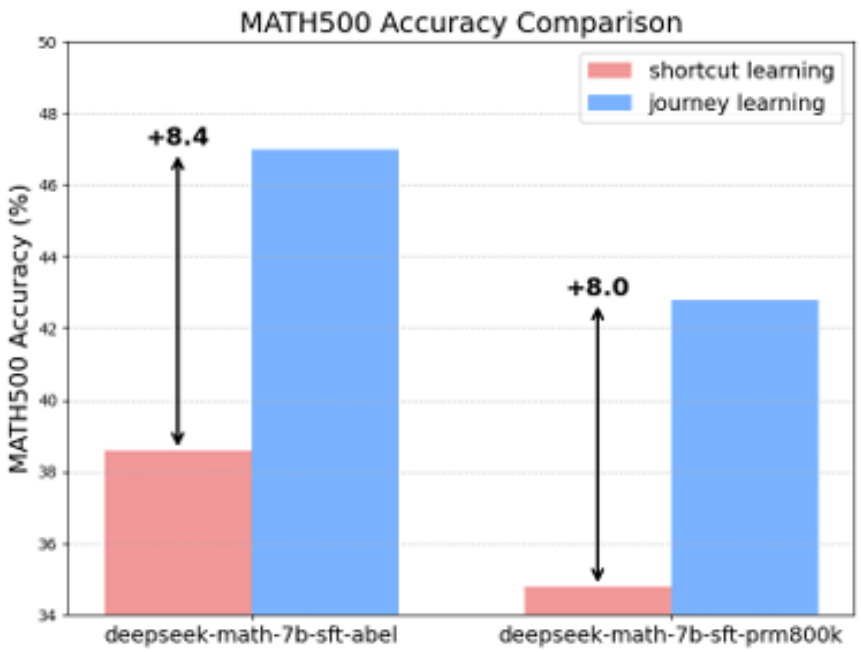
[插入 Verifier 的回馈]

2.4.2 生成推理过程的每一步都必须正确吗？

https://arxiv.org/abs/2410.18982

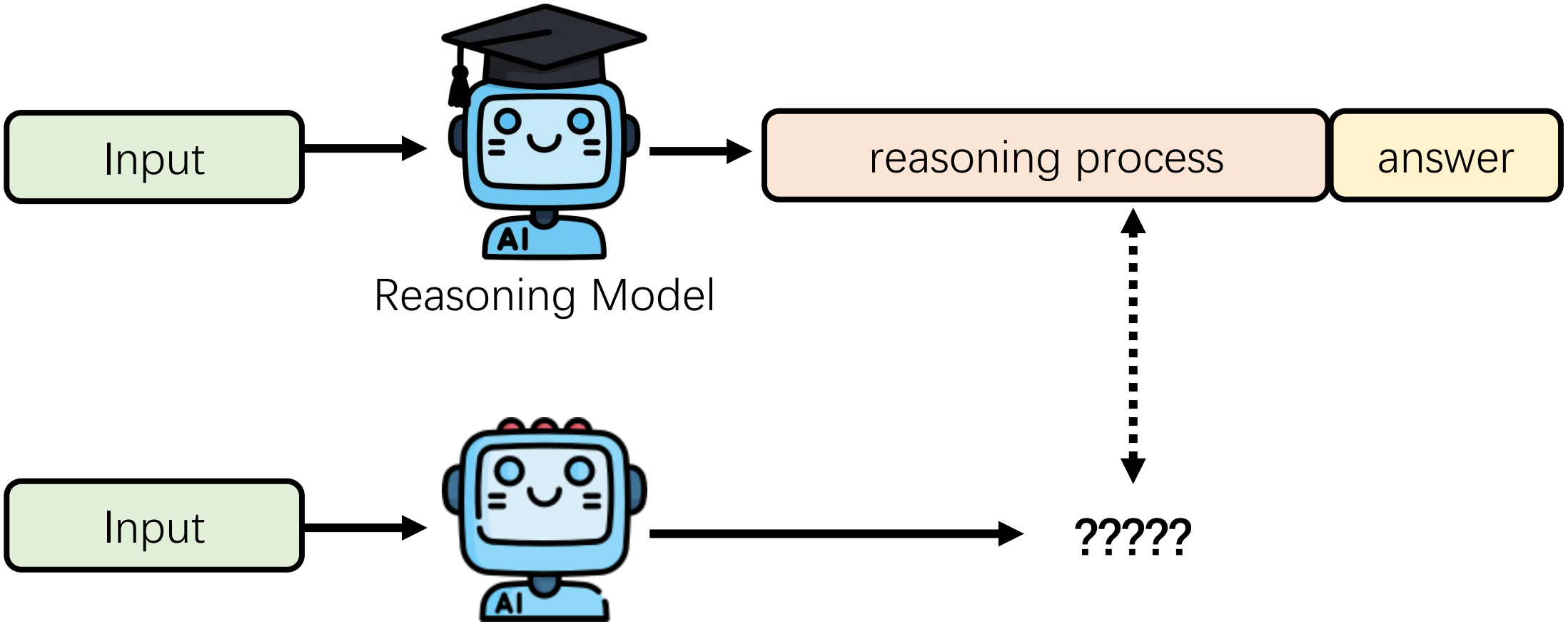


(a) Shortcut learning. (b) Journey learning



(c) Performance Comparison

2.4.3 更方便的知识蒸馏方法



Sky-T1: https://novasky-ai.github.io/posts/sky-t1/
s1:https://arxiv.org/abs/2501.19393

2.4.3 更方便的知识蒸馏方法

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCode Bench	CodeForces
	pass@1	cons@64	pass@1	pass@1	pass@1	rating
GPT-4o-0513	9.3	13.4	74.6	49.9	32.9	759
Claude-3.5-Sonnet-1022	16.0	26.7	78.3	65.0	38.9	717
OpenAI-o1-mini	63.6	80.0	90.0	60.0	53.8	**1820**
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9	1316
DeepSeek-R1-Distill-Qwen-1.5B	28.9	52.7	83.9	33.8	16.9	954
DeepSeek-R1-Distill-Qwen-7B	55.5	83.3	92.8	49.1	37.6	1189
DeepSeek-R1-Distill-Qwen-14B	69.7	80.0	93.9	59.1	53.1	1481
DeepSeek-R1-Distill-Qwen-32B	**72.6**	83.3	94.3	62.1	57.2	1691
DeepSeek-R1-Distill-Llama-8B	50.4	80.0	89.1	49.0	39.6	1205
DeepSeek-R1-Distill-Llama-70B	70.0	**86.7**	**94.5**	**65.2**	**57.5**	1633

Foundation Model

2.4 以结果导向的推理

更强的思维链（Chain-of-Thought, CoT）

设计模型推理的流程

教模型推理过程（Imitation Learning）

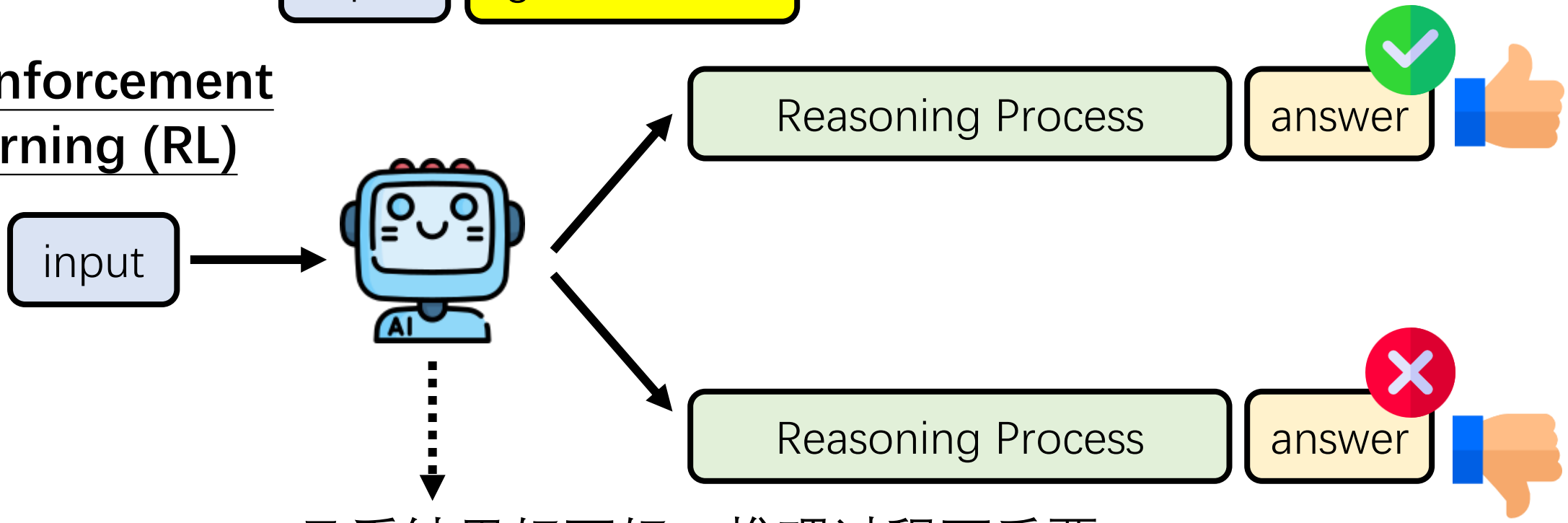
以结果导向的推理（Reinforcement Learning, RL）

2.4 以结果导向的推理

https://arxiv.org/abs/2501.12948

Training Data: input | ground truth

Reinforcement Learning (RL)



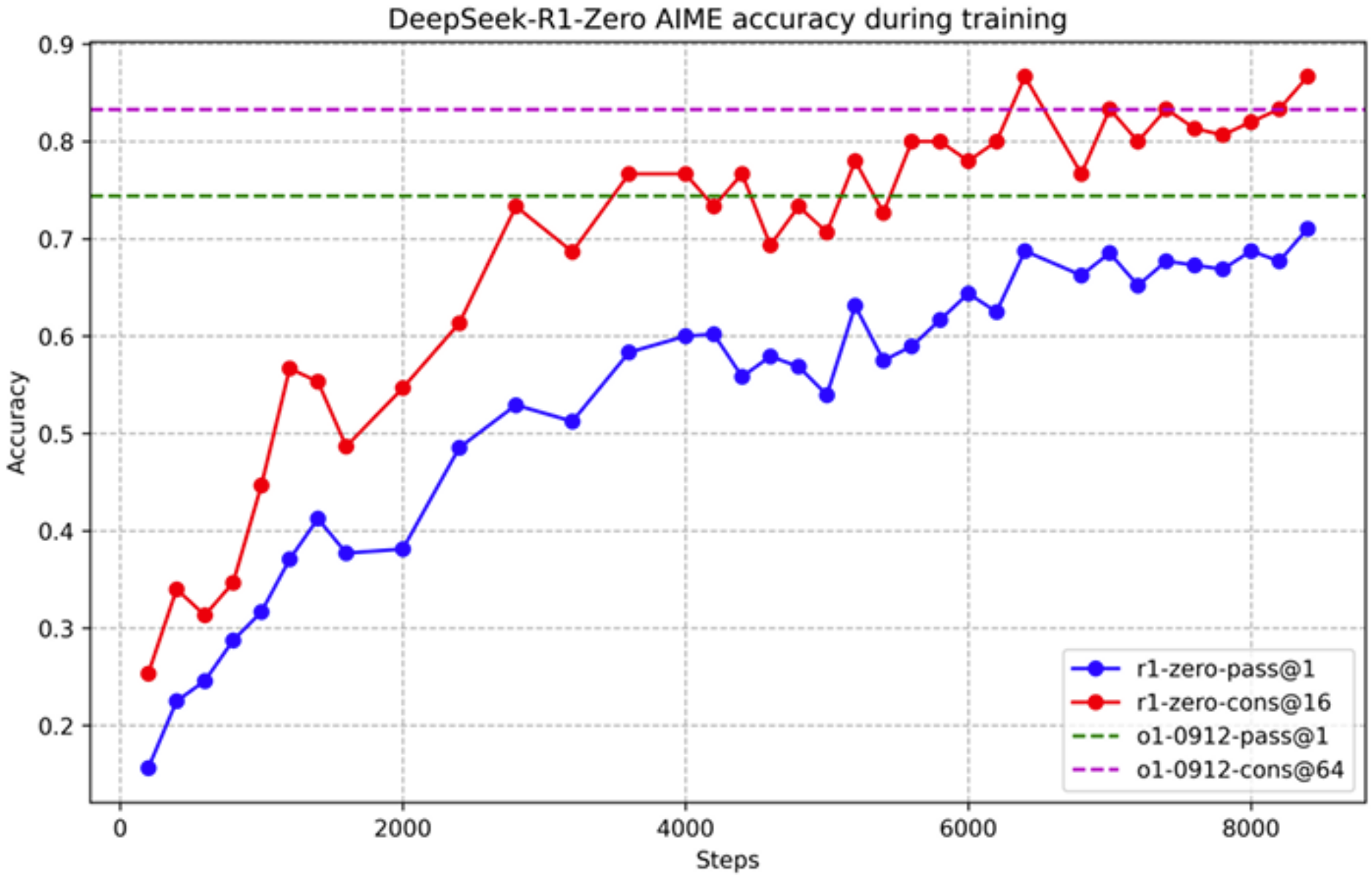
只看结果好不好，推理过程不重要

DeepSeek-v3-base (Foundation Model) —RL→ DeepSeek-R1-Zero

Accuracy/Think as reward

2.4 以结果导向的推理

https://arxiv.org/abs/2501.12948



https://arxiv.org/abs/2501.12948

Aha Moment

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

$\cdots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a + x}} = x$

First, let's square both sides:

$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

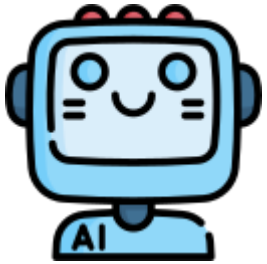
$\cdots$

2.4 以结果导向的推理

https://arxiv.org/abs/2501.12948

Training Data: input | ground truth

Reinforcement Learning (RL)

input → 

→ Reasoning Process | answer

→ Reasoning Process | answer

只看结果好不好，推理过程不重要

DeepSeek-v3-base (Foundation Model) —RL, Accuracy/Think as reward→ DeepSeek-R1-Zero

Unreadable

2.5 DeepSeek-V3 到 DeepSeek-R1

DeepSeek-v3-base → RL → DeepSeek-R1-Zero

Accuracy/Think as reward

Input | reasoning process | ground truth

- using few-shot prompting with a long CoT as an example
- directly prompting models to generate detailed answers with reflection and verification

Generated data + **human annotation**

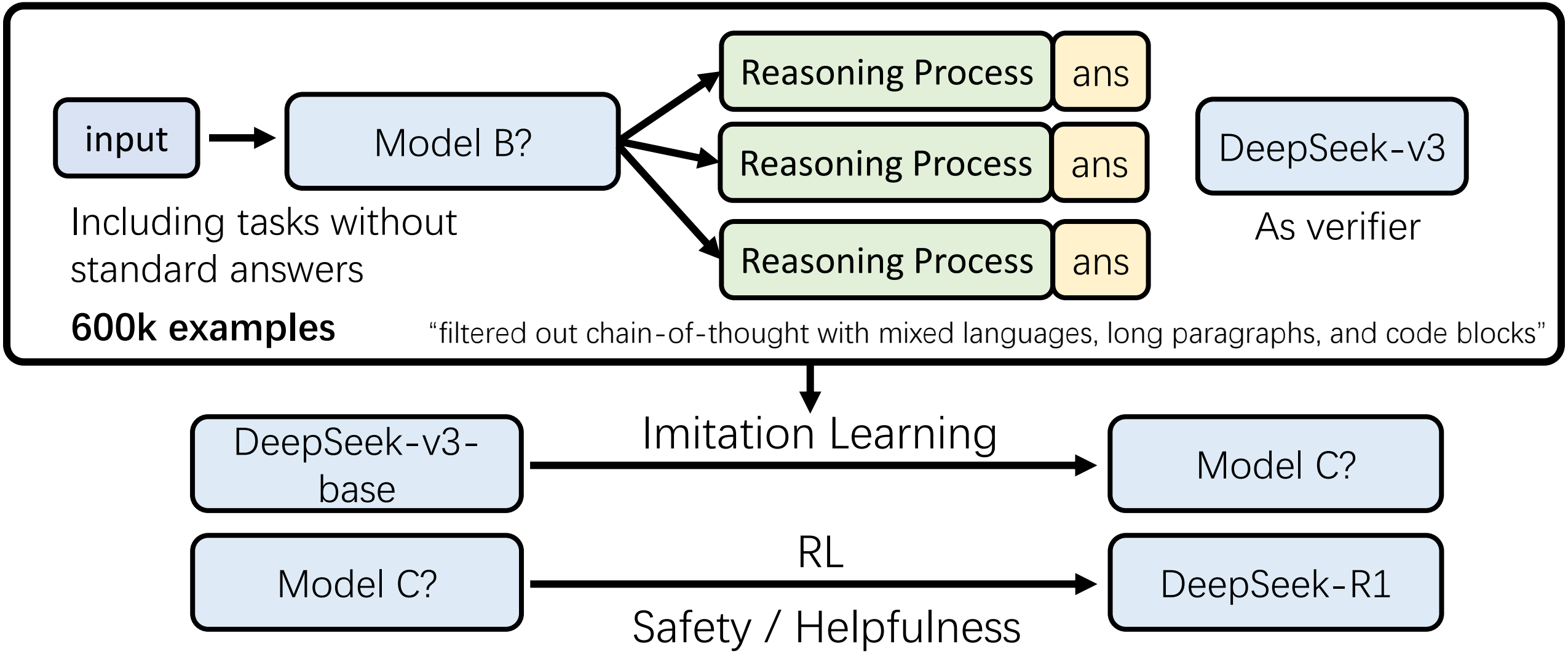
(**Thousands of examples**)

DeepSeek-v3-base → Imitation Learning → Model A?

Model A? → RL → Model B?

Accuracy / **Language coherence** as reward

2.5 DeepSeek-V3 到 DeepSeek-R1



Based on the Deepseek-R1 paper, both the process verifier and MTCS were tried but ultimately not used.

2.5 DeepSeek-V3 到 DeepSeek-R1

使用Mermaid绘制一个图书管理系统的用例图。参与者包括用户、图书馆管理员、系统管理员。功能包括人员管理、图书管理、图书预订管理、用户行为分析等，但是这些功能需要稍微细化一下。

已深度思考（用时 174 秒）

DeepSeek-R1的推理过程几乎是自动的，所以可读性还是存在问题，但不影响推理的结果

题目中只提到了“图书预订管理”，但没有说借还。所以，我可能只需要考虑预订。但为了完整，或许应该包括基本的借还功能？用户行为分析可能涉及借阅行为，所以借还应该是存在的。

再读题目："功能包括人员管理、图书管理、图书预订管理、用户行为分析等"，所以它列出了这些，但“等”表示可能还有其他。但为了简化，我应该只基于给定的功能细化。

但图书管理可能包括借还操作？预订管理是单独的。用户行为分析可能基于借阅历史。

或许我需要包括借还用例。

2.6 推理模型的能力依赖Base Model的能力

以 Qwen-32B-Base 作为 Base Model

Model	AIME 2024		MATH-500	GPQA Diamond	LiveCodeBench
	pass@1	cons@64	pass@1	pass@1	pass@1
QwQ-32B-Preview	50.0	60.0	90.6	54.5	41.9
DeepSeek-R1-Zero-Qwen-32B	47.0	60.0	91.6	55.0	40.2
DeepSeek-R1-Distill-Qwen-32B	**72.6**	**83.3**	**94.3**	**62.1**	**57.2**

- RL 无法大幅强化 Qwen-32B-Base 的能力
- Imitation Learning (向 DeepSeek-R1 学习)对Qwen-32B-Base反而比较有效

RL强化模型原有的能力

2.6 推理模型的能力依赖Base Model的能力

4.5 如何避免灾难性遗忘?

(Pseuc

Parap

inpu

也许RLHF可以更好的避免灾难性遗忘

RLHF
基于人类反馈的强化学习

d

Self-output

input: new → Foundation Model → output: old

output: new

If it is correct, use this.

2.6 推理模型的能力依赖Base Model的能力

Base Model
本来就会 Aha

Question(a): Find the least positive integer n such that $1/\sin 45^\circ \sin 46^\circ + 1/\sin 47^\circ \sin 48^\circ + \cdots + 1/\sin 133^\circ \sin 134^\circ = 1/\sin n^\circ$.

Response

First, I notice that the left-hand side of the equation has the form $\sum_{k=1}^{n} \frac{1}{\sin a_k \sin a_{k+1}}$, where $a_1 = 45^\circ, a_2 = 47^\circ, \ldots, a_{90} = 133^\circ$. I also notice that $\sin a_{k+1} = \sin(a_k + 2^\circ)$.

...

Simplifying, I get $\sin(3a) = 3\sin a - 4\sin^3 a$.

Aha! I can use this to get $\sin(a + 2b) = \sin((a + b) + b) = 3\sin(a + b)$ ...

This looks messy, but I notice that it has the form $\sin a \cos b + \cos a \sin b$ again.

This gives me an idea. Maybe I can use this identity repeatedly to simplify the left-hand side of the equation.

...

Question(b): An element is randomly chosen from among the first 15 rows ...

Response

In Pascal's Triangle, every row starts and ends with 1, ...

...

This can be calculated as: *awkward silence* Wait, I'm overthinking. Let's try again. The number of elements in the first n rows of Pascal's Triangle is $(n + 1)!/[2! * (n - 1)!]$.

...

Question(c): Denali and Nate work for a dog walking business and are paid ...

Response

To solve this problem, let's start by defining the variables...

...

Again, since $x \neq 0$, this equation doesn't give us a valid solution either. Let's check if we made an error. We should verify the problem conditions again. The correct approach is to ...

...

DeepSeek-v3

https://arxiv.org/abs/2503.20783

推理过程不是越长越好

3.1 推理过程长就一定好吗？

计算123*456

word count: 4

已深度思考 (1m 25s)

首先，用户要求计算123乘以456。这是一个简单的乘法问题，我需要准确地计算它。

让我回想一下乘法的基本步骤：将两个数相乘，我可以使用长乘法的方法。或者，我可以直接计算，因为数字不大。

123乘以456。我可以分解它来更容易计算。

3.1 推理过程长就一定好吗？

第一次计算

让我计算每个部分：

首先，123 * 400。123 * 4 = 492，然后乘以100（因为400是4*100），所以492 * 100 = 49200。

实际上，123 * 400：123 * 4 = 492，然后加两个零，所以49200。

其次，123 * 50。123 * 5 = 615，然后乘以10（因为50是5*10），所以615 * 10 = 6150。

第三，123 * 6。123 * 6 = 738。

现在，把它们加起来：49200 + 6150 + 738。

先加49200和6150。49200 + 6000 = 55200，再加150 = 55350。

或者，49200 + 6150：49200 + 6000 = 55200，然后55200 + 150 = 55350。

现在加738：55350 + 738。

55350 + 700 = 56050，然后56050 + 38 = 56088。

第一次验证

我应该验证一下我的计算。让我用另一种方法计算。

我可以直接计算123 * 456。

使用长乘法：

首先，写下123和456。

乘以个位数：6 * 123。

⋮

所以，结果是56088。

是的，和我之前计算的一致：56088。

3.1 推理过程长就一定好吗？

第二次验证

我可以使用计算器验证，但既然我是AI，我应该确保正确。

123 * 456。

我可以计算100*456=45600，20*456=9120，3*456=1368。

然后加：45600 + 9120 = 54720，然后54720 + 1368。

54720 + 1300 = 56020，然后56020 + 68 = 56088。

相同结果。

第三次验证

或者，123 * 456。

456 * 120 + 456 * 3 = 456*100=45600，456*20=9120，所以456*120=45600+9120=54720。

然后456*3=1368。

54720+1368=56088。

所以，确认结果是56088。

第四次验证

让我心算：120*450=54000，但更精确些。

123*400=49200，123*56=123*50=6150，123*6=738，然后49200+6150=55350，55350+738=56088。是的。

3.1 推理过程长就一定好吗？

第五次验证

或者，456*123。

456*100=45600

456*20=9120

456*3=1368

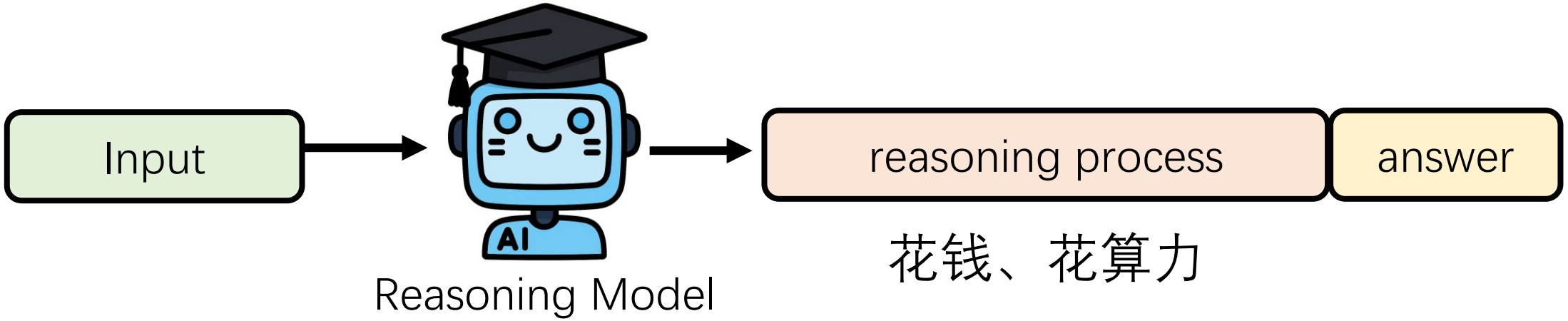
总和：45600+9120=54720；54720+1368。

54720+1000=55720；55720+300=56020；56020+68=56088。

完美。

所以，答案是56088。

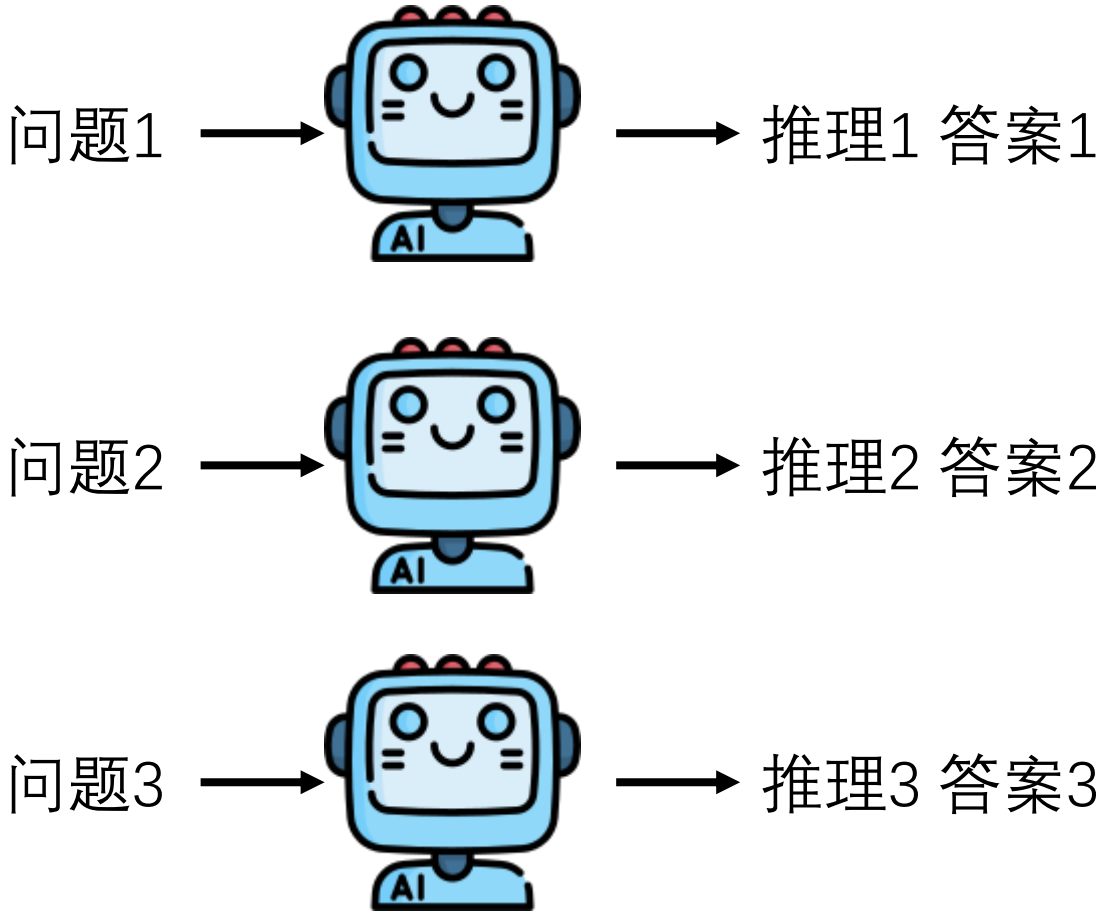
3.1 推理过程长就一定好吗？

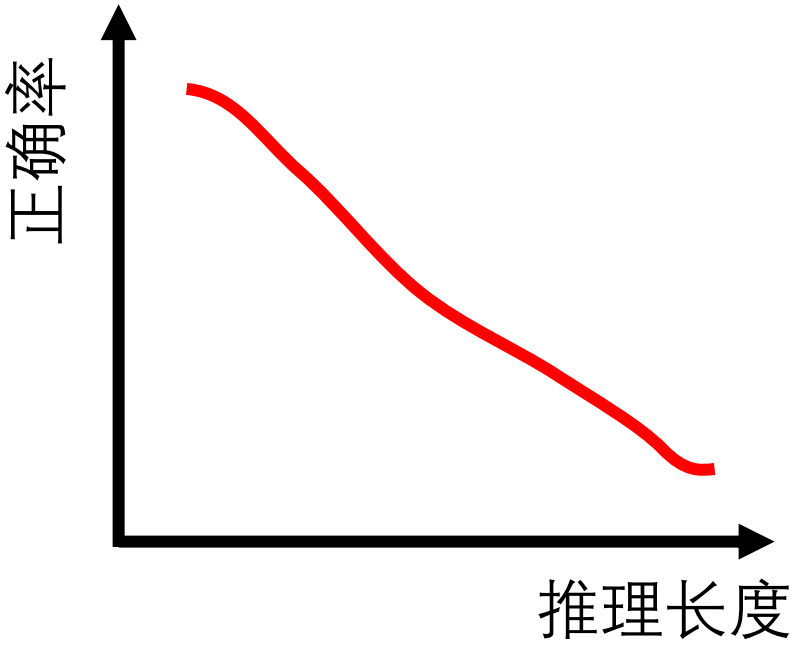


花钱、花算力

可不可以缩短推理过程？

3.1 推理过程长就一定好吗？

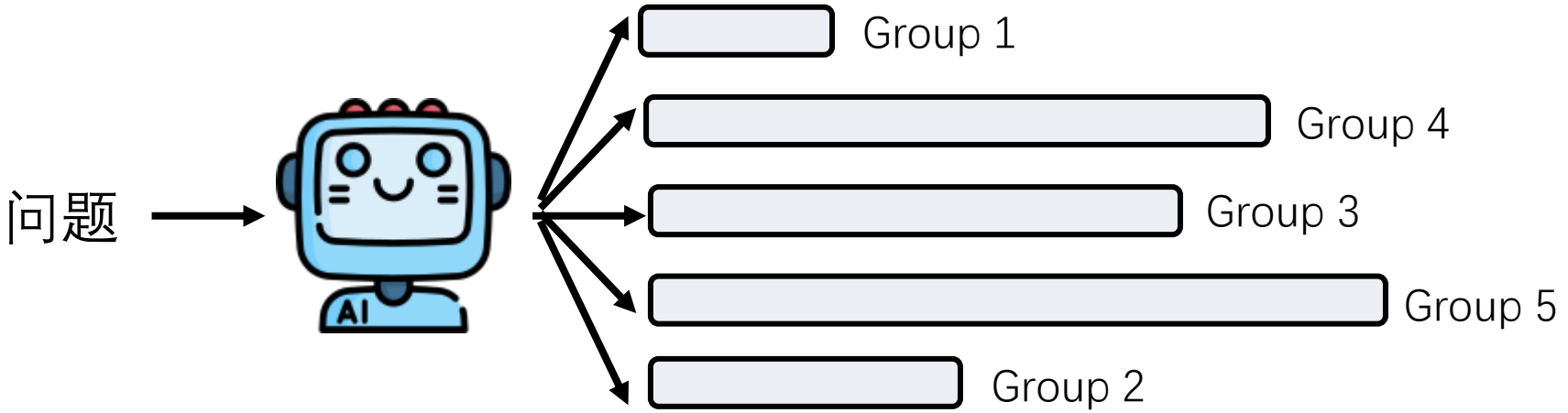


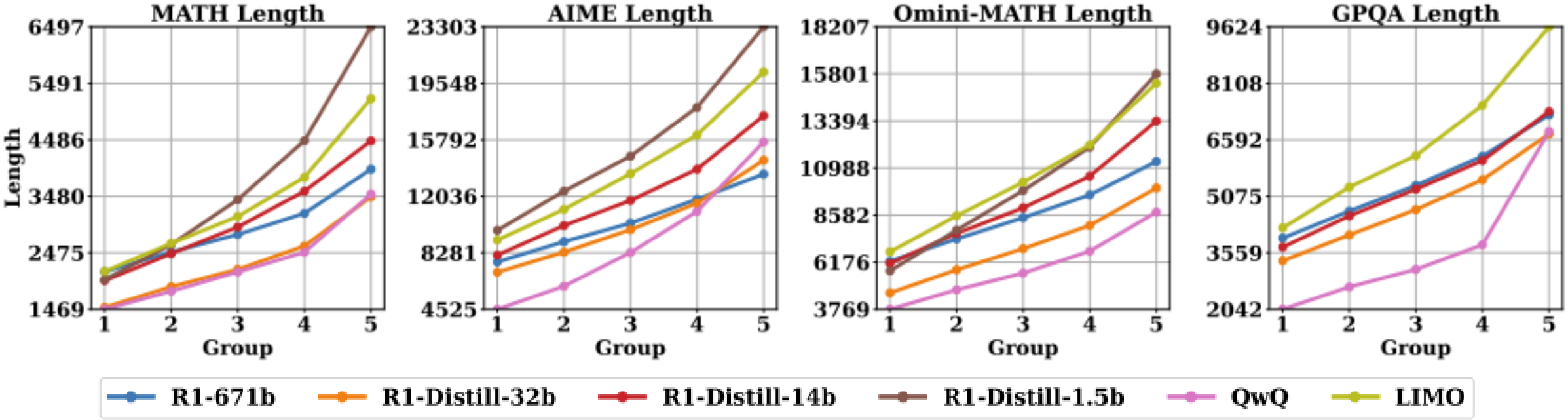


问题难度↑ → 正确率↓

问题难度↑ → 推理长度↑

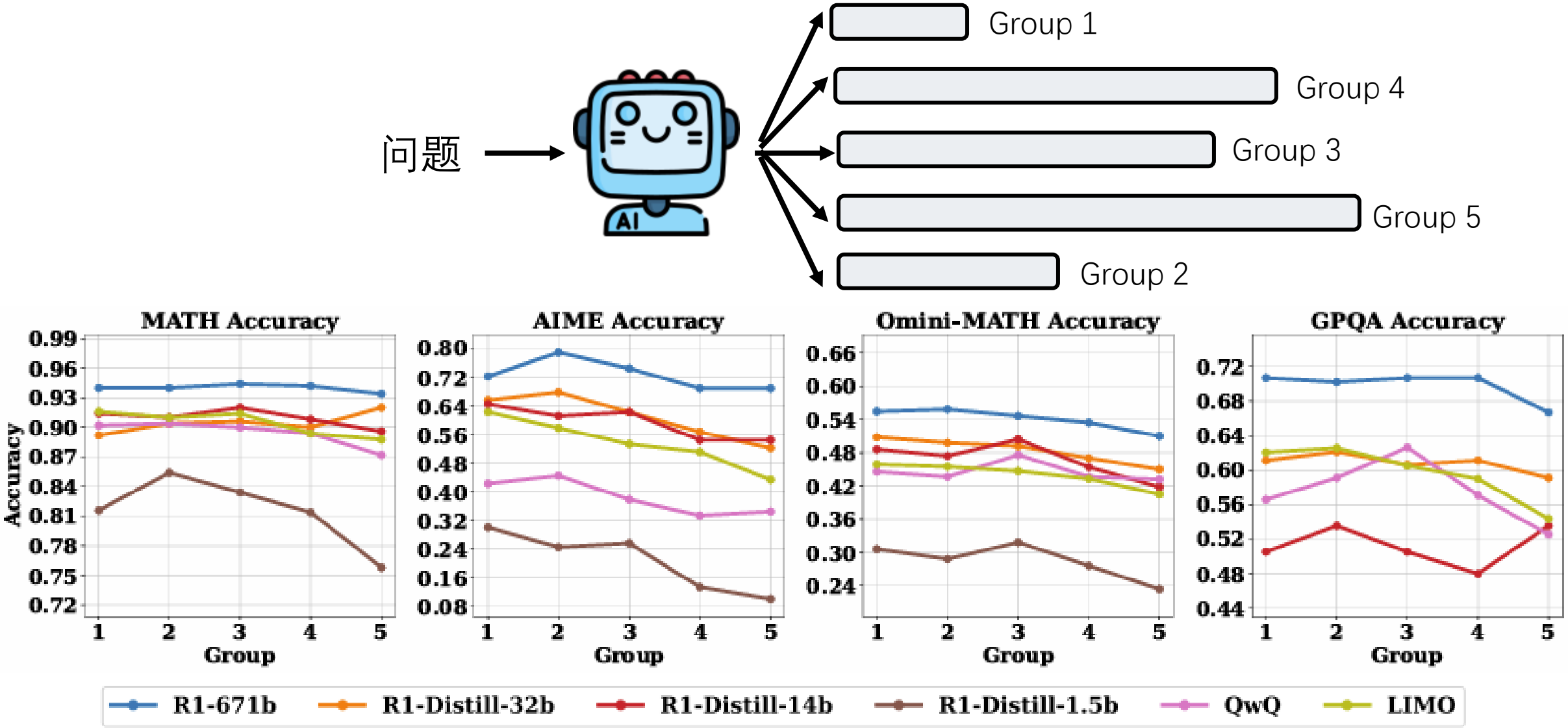
3.1 推理过程长就一定好吗？





https://arxiv.org/abs/2502.12215

3.1 推理过程长就一定好吗？



https://arxiv.org/abs/2502.12215

3.2 如何缩短推理长度?

更强的思维链（Chain-of-Thought, CoT）

设计模型推理的流程

教模型推理过程（Imitation Learning）

以结果导向的推理（Reinforcement Learning, RL）

3.2.1 控制思维链的长度

Chain of Draft

https://arxiv. org/pdf/2502.18600

Standard

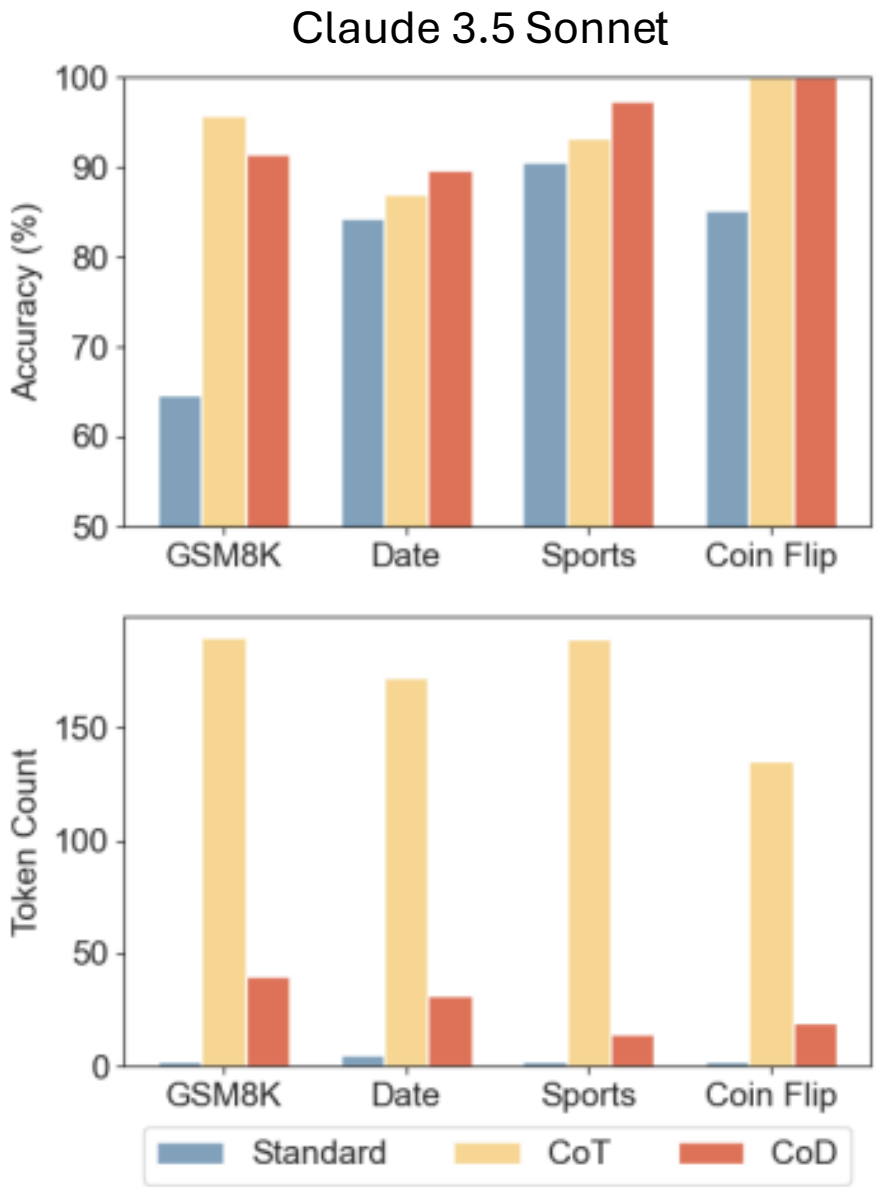
```
Answer the question directly. Do not return any
preamble, explanation, or reasoning.
```

Chain-of-Thought

```
Think step by step to answer the following question.
Return the answer at the end of the response after a
separator ####.
```

Chain-of-Draft

```
Think step by step, but only keep a minimum draft for
each thinking step, with 5 words at most. Return the
answer at the end of the response after a separator
####.
```



3.2.2 降低推理流程的复杂度

更强的思维链（Chain-of-Thought, CoT）

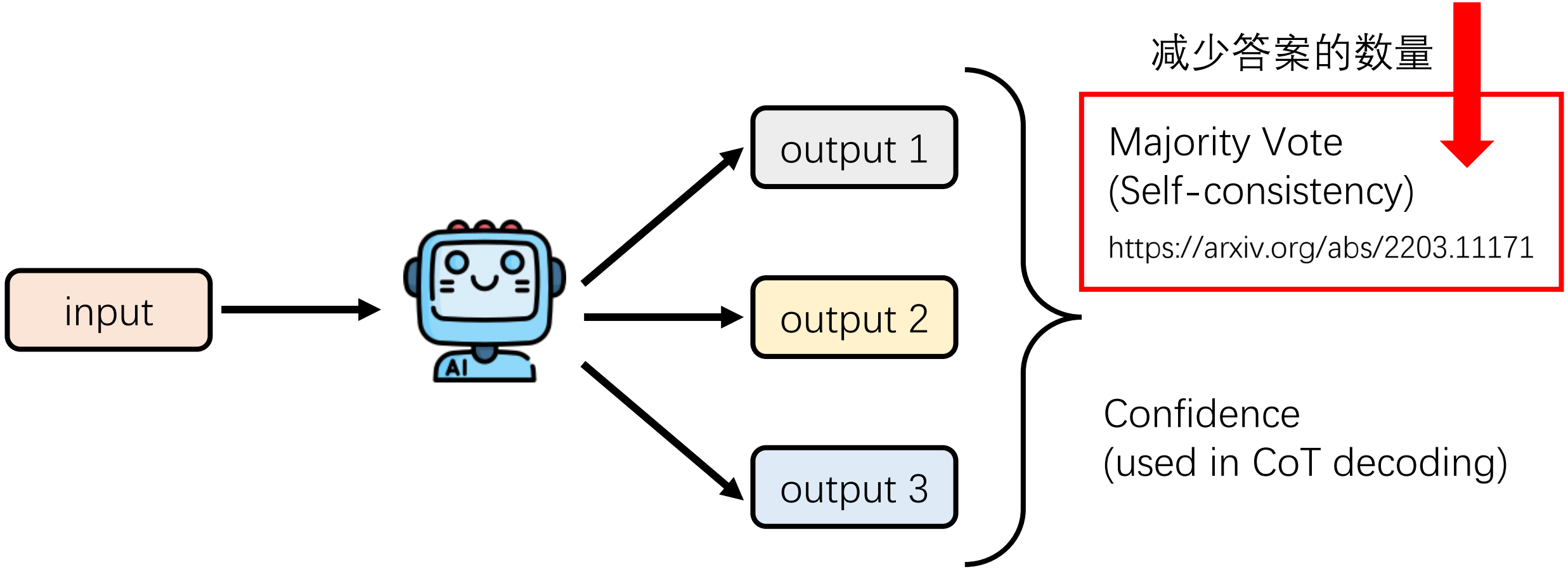
设计模型推理的流程

教模型推理过程（Imitation Learning）

以结果导向的推理（Reinforcement Learning, RL）

3.2.2 降低推理流程的复杂度

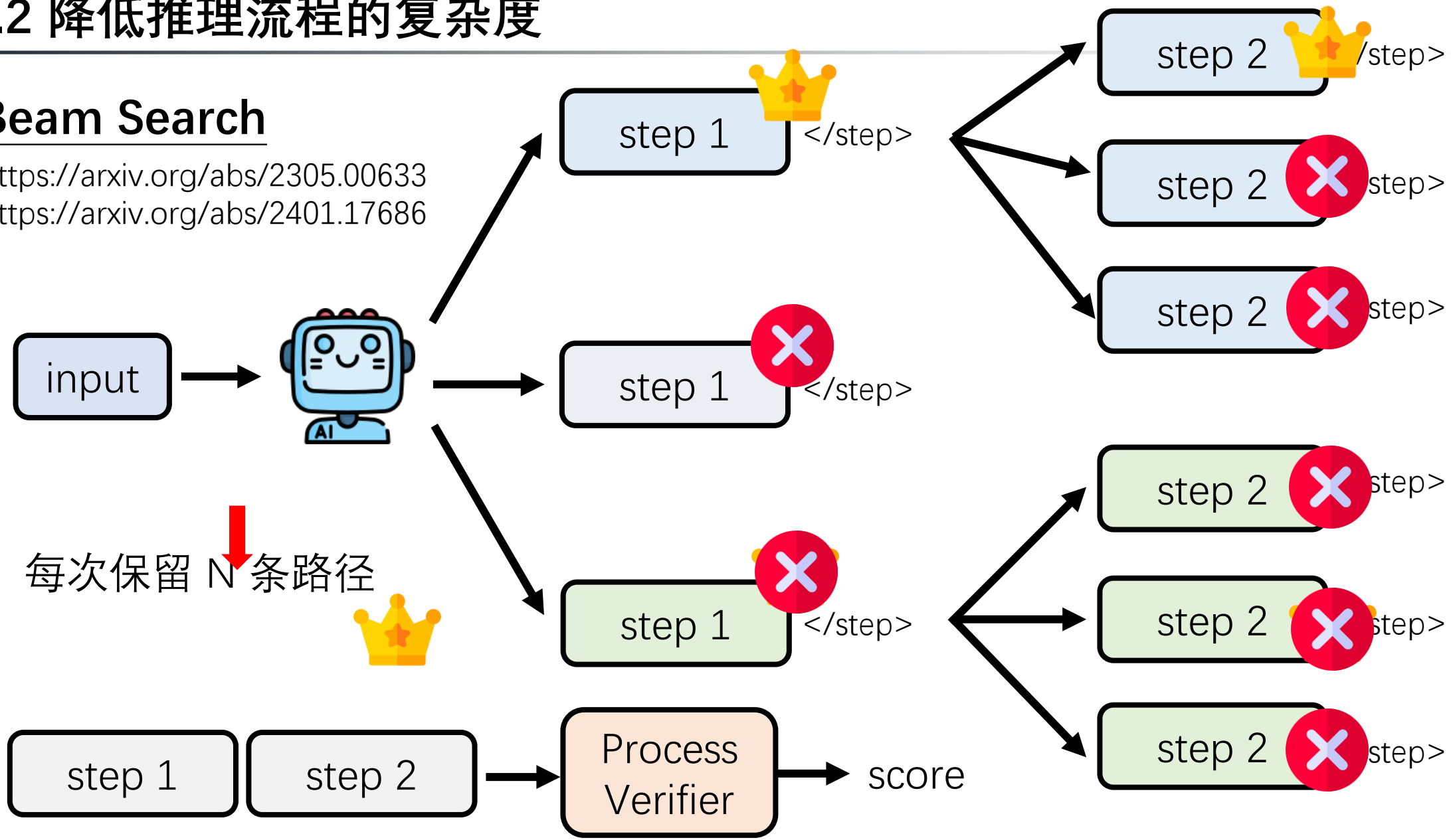
- 让LLMs少试几次



3.2.2 降低推理流程的复杂度

Beam Search

https://arxiv.org/abs/2305.00633
https://arxiv.org/abs/2401.17686



3.2.3 教模型简短的推理过程

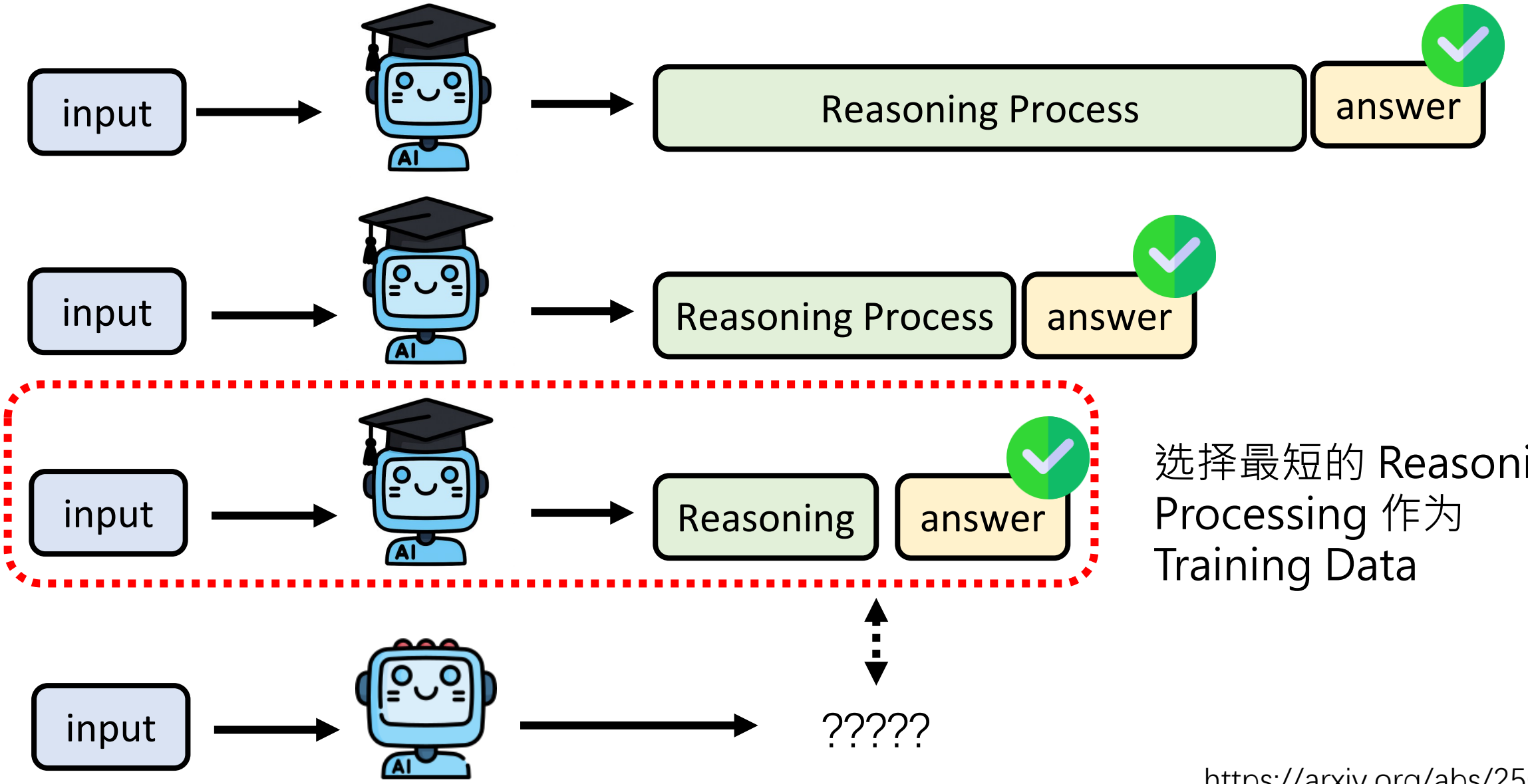
更强的思维链（Chain-of-Thought, CoT）

设计模型推理的流程

教模型推理过程（Imitation Learning）

以结果导向的推理（Reinforcement Learning, RL）

3.2.3 教模型简短的推理过程



选择最短的 Reasoning Processing 作为 Training Data

https://arxiv.org/abs/2502.18080

From Explicit CoT to Implicit CoT

		Input	CoT	Output
Explicit CoT	Stage 0:	2 1 × 4 3 =	8 4 + 0 6 3	= 8 0 4
	Stage 1:	2 1 × 4 3 =	4 + 0 6 3	= 8 0 4
	Stage 2:	2 1 × 4 3 =	+ 0 6 3	= 8 0 4
	Stage 3:	2 1 × 4 3 =	0 6 3	= 8 0 4
	Stage 4:	2 1 × 4 3 =	6 3	= 8 0 4
	Stage 5:	2 1 × 4 3 =	3	= 8 0 4
Implicit CoT	Stage 6:	2 1 × 4 3 =		= 8 0 4

只适合较为简单的问题

https://arxiv.org/abs/2405.14838

3.2.4 将长度作为结果的评测标准之一

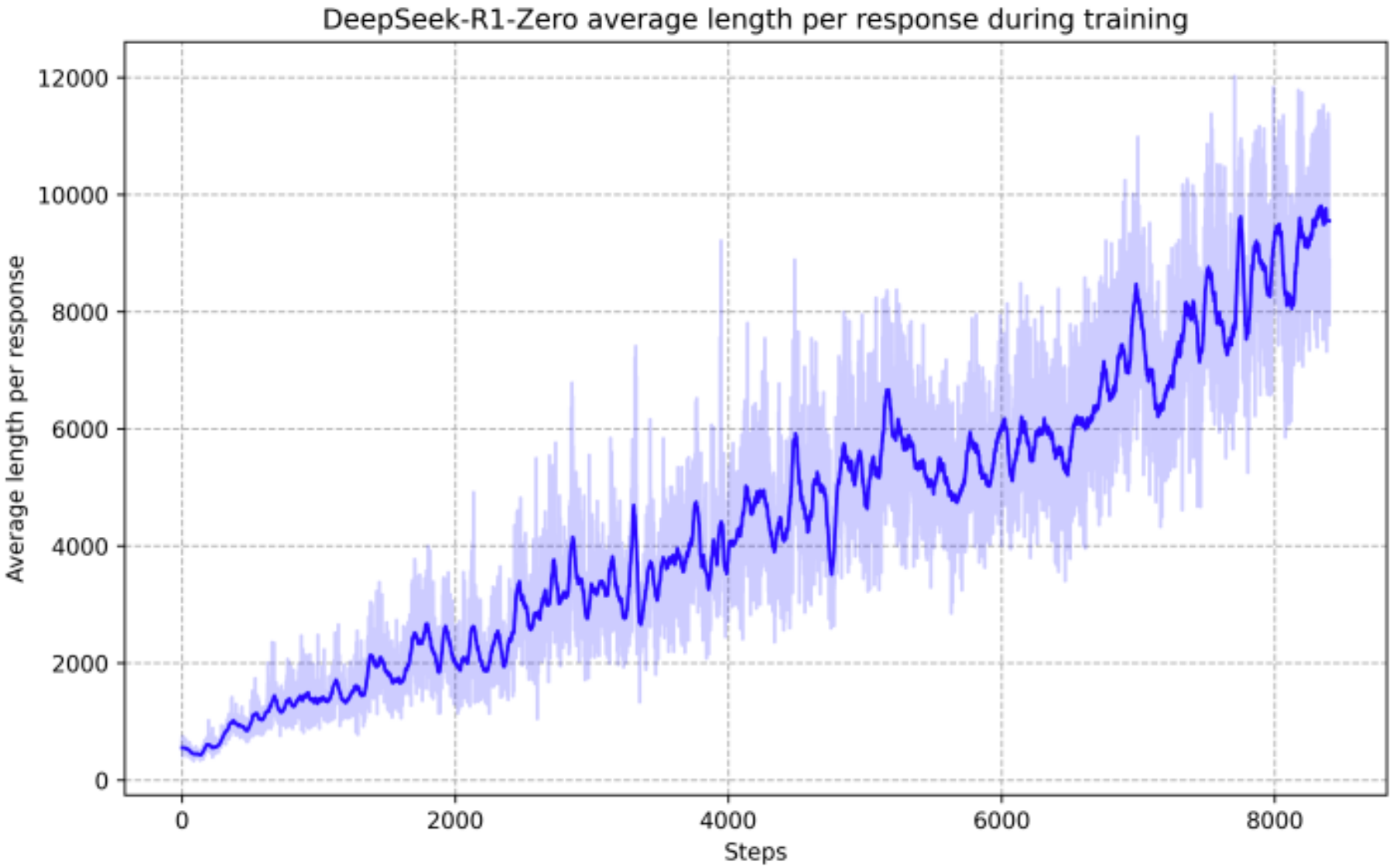
更强的思维链（Chain-of-Thought, CoT）

设计模型推理的流程

教模型推理过程（Imitation Learning）

以结果导向的推理（Reinforcement Learning, RL）

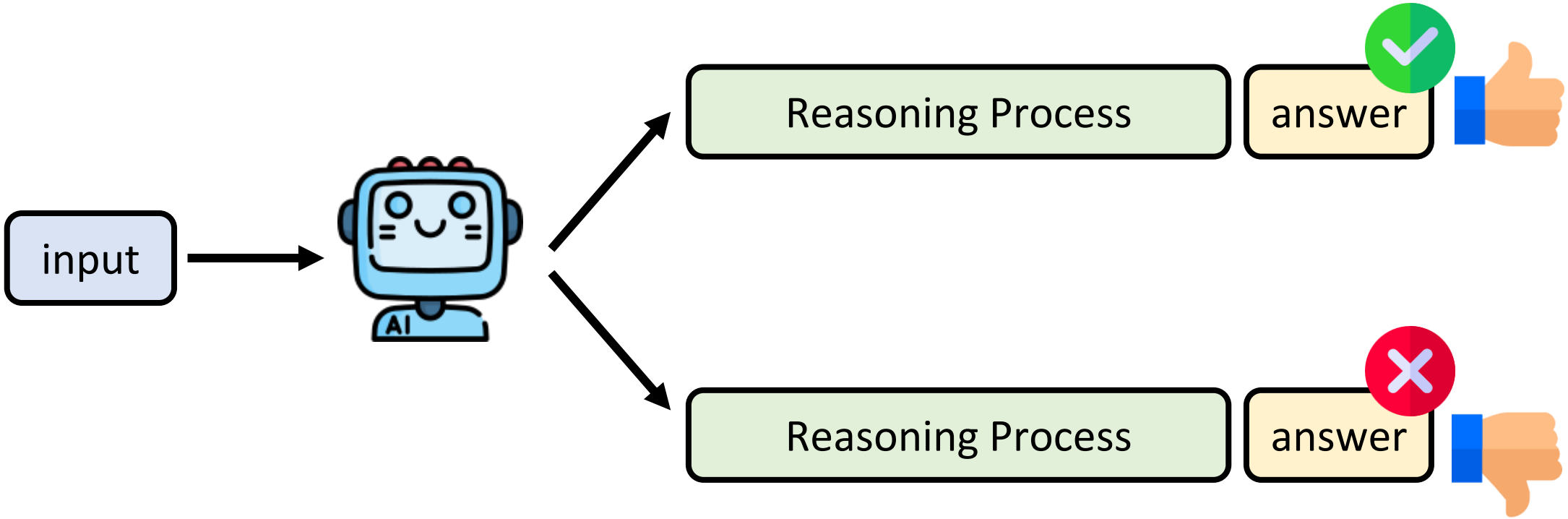
3.2.4 将长度作为结果的评测标准之一



DeepSeek-R1

https://arxiv.org/abs/2501.12948

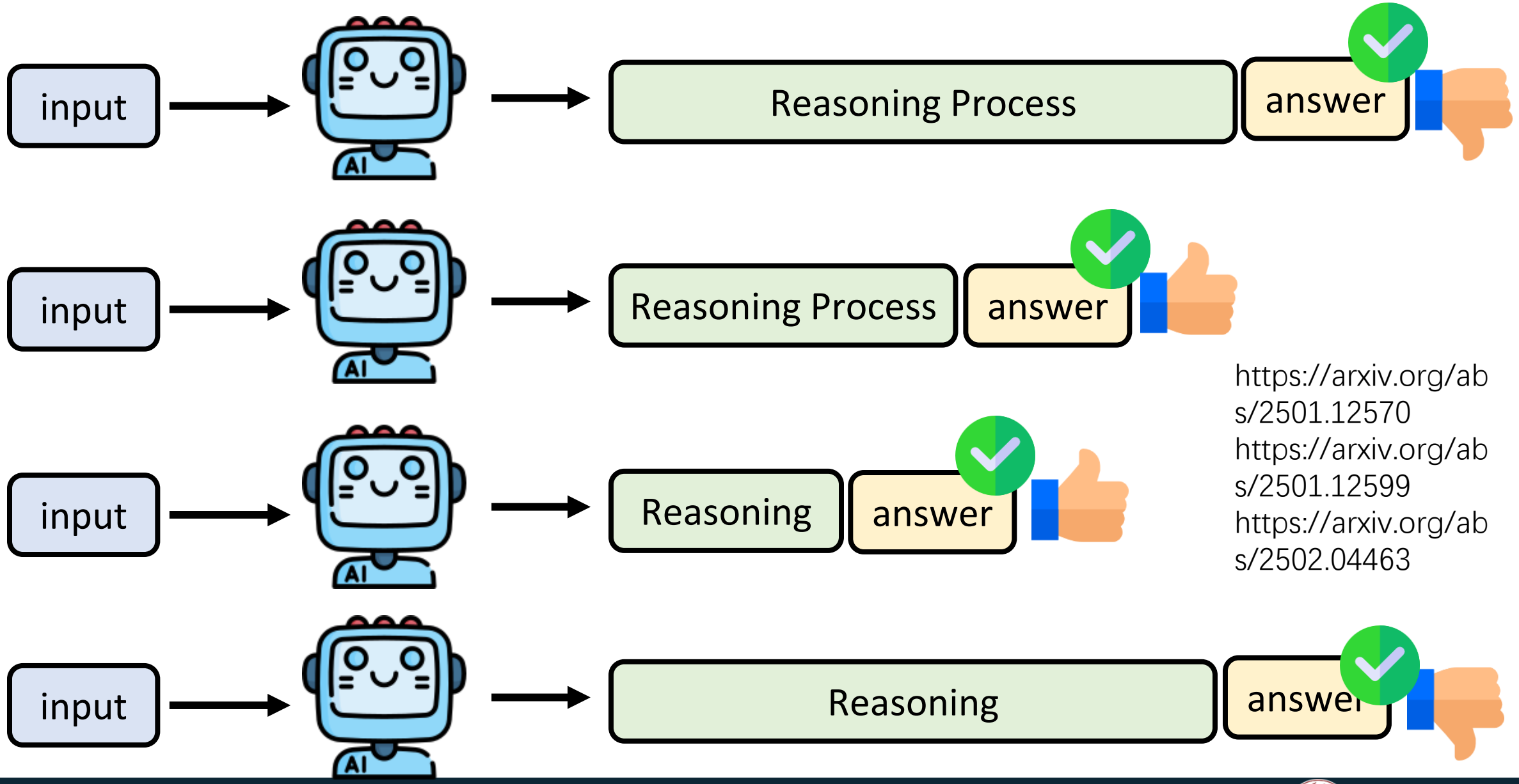
3.2.4 将长度作为结果的评测标准之一



没人教 AI 要考虑推理过程的长度!

超过一定长度就算不好的？

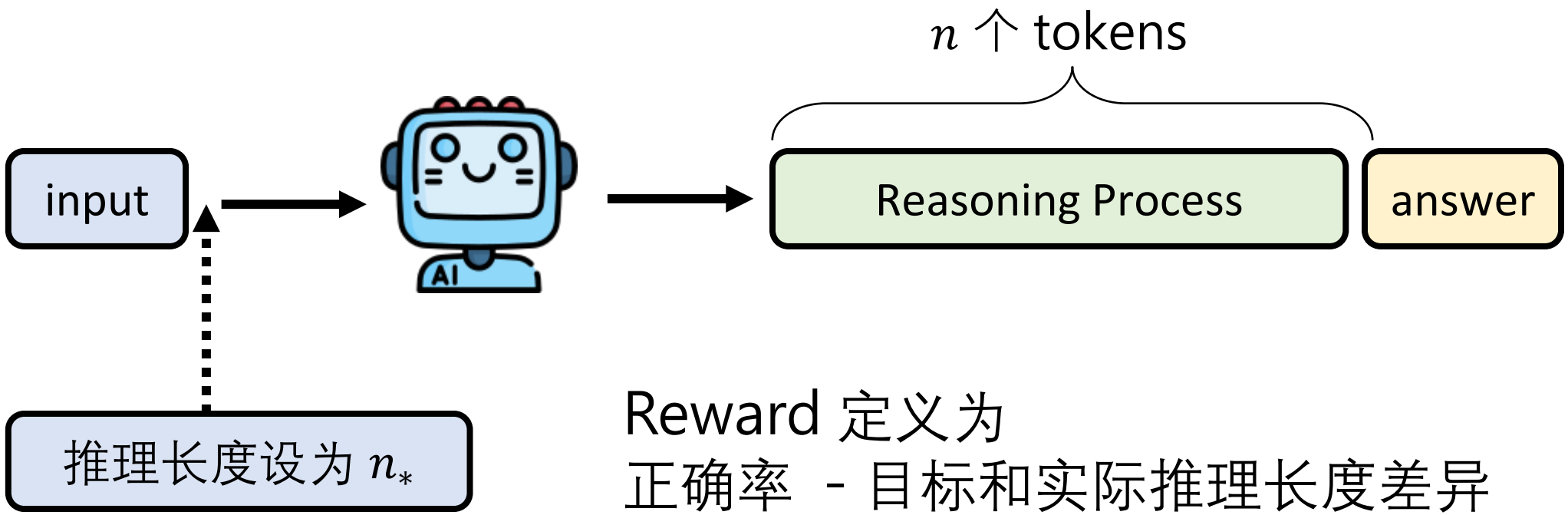
3.2.4 将长度作为结果的评测标准之一



https://arxiv.org/abs/2501.12570
https://arxiv.org/abs/2501.12599
https://arxiv.org/abs/2502.04463

3.2.4 将长度作为结果的评测标准之一

- 控制模型的推理长度



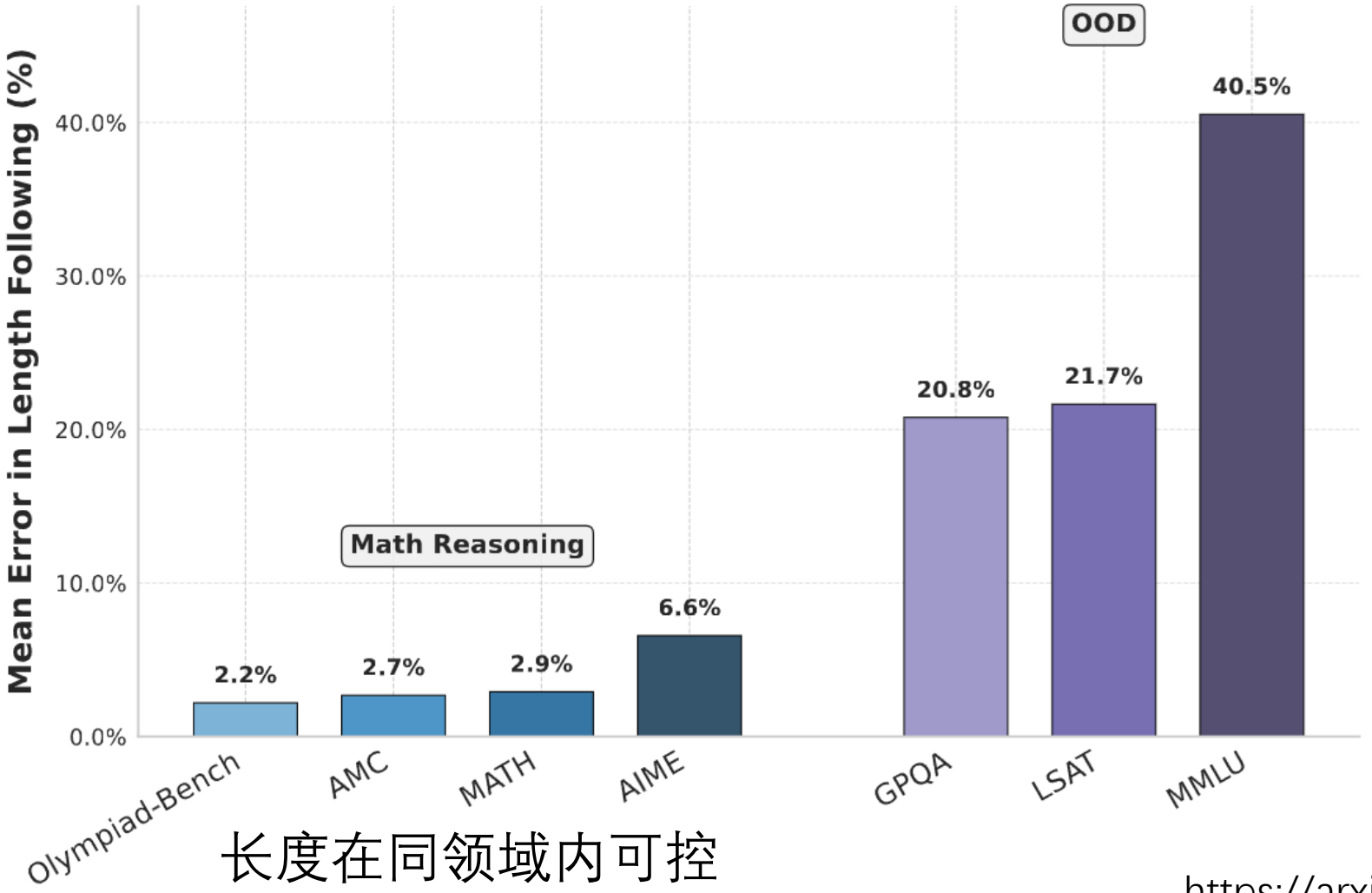
Reward 定义为
正确率 - 目标和实际推理长度差异

https://arxiv.org/abs/2503.04697

3.2.4 将长度作为结果的评测标准之一

- 控制模型的推理长度

长度在不同领域内不完全可控，但有效果

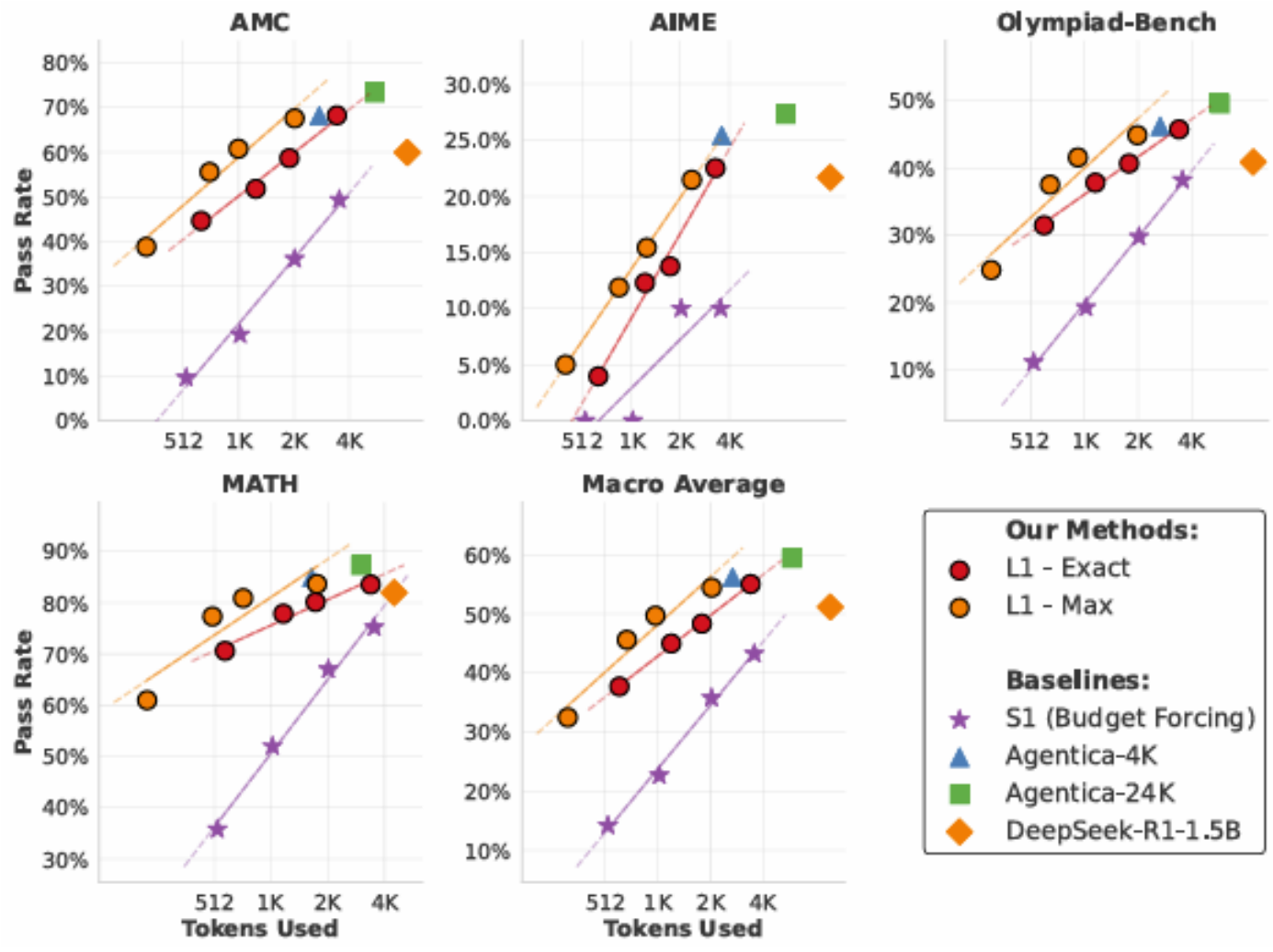


长度在同领域内可控

https://arxiv.org/abs/2503.04697

3.2.4 将长度作为结果的评测标准之一

- 控制模型的推理长度



L1-Exact：准确控制

L1-Mac：比目标短

用RL的方式控制长度比粗暴的方式（S1）更有效

https://arxiv.org/abs/2503.04697

让模型推理的四种方法	控制模型推理长度的四种方法
更强的思维链	控制思维链长度
设计模型推理的流程	降低流程复杂度
教模型推理过程	简短的推理过程
以结果导向的推理	将长度作为结果